

Observer-Patch Holography and the Dark Matter Phenomenon

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Abstract

OPH models observer-like self-reading systems as bounded patches with local state, overlap boundaries, readback records, repair moves, and public receipts. Its implemented static galaxy equation is QUMOND with the empirical McGaugh–Lelli–Schombert interpolation. The collar-count construction represents that interpolation, but the flux-recovery closure is a physical premise, and static data identify only $a_{0,\text{eff}} = a_{0,\text{OPH}}/\lambda_{\text{collar}}^2$. Universal application of the static law is excluded by the Cassini quadrupole constraint. A zero-flux cut does not remove external tides and is incompatible with disk fields. The QUMOND virtual density can be signed, so it supplies neither a positive transported stress nor a relativistic lensing prediction. The OPH-R1 completion contract specifies one observer-foliated retarded response with an exact QUMOND static limit, causal propagation, and recipient stress for damping. Its microphysical kernel, covariant action, lensing, cluster, and cosmological likelihoods are work in progress; the paper does not claim a physical dark-matter solution.

What This Paper Contributes

The static galaxy equation in this paper is QUMOND with the empirical RAR interpolation family. Its Poisson-count representation is OPH-motivated, while the flux-recovery closure is a premise. Every static prediction depends only on

$$a_{0,\text{eff}} = a_{0,\text{OPH}}/\lambda_{\text{collar}}^2,$$

so galaxy data cannot validate $a_{0,\text{OPH}}$ and λ_{collar} separately.

The universal QUMOND continuation is excluded by Cassini. The closed-cut predicate is invalid: zero net flux does not imply zero tide, and enclosed-mass activation omits the exterior disk mass used by the SPARC field calculation. It supplies no Solar-System, wide-binary, satellite, or disk prediction.

The constructive object is OPH-R1: one observer-foliated, retarded response functional with the QUMOND static limit, causal propagation, and total-stress closure through an explicit recipient sector. It contains no system-specific switch or internal/external source label. Derivation of its kernel from finite collar readback and repair is work in progress.

The paper separates the finite collar and modular-source theorems, the static QUMOND continuation, the covariant response parent required for relativistic observables, and the empirical tests of that continuation. None of the first three objects by itself constitutes a physical dark-matter solution.

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1 Plain Picture

Finite record repair may contribute to gravity, but the required stress is not derived from the collar source. The derivation is work in progress.

At equilibrium, the candidate response has to reproduce a conservative galaxy potential. Away from equilibrium, the same parent has to propagate and relax causally, generate lensing and cosmological stress, and conserve total stress. All baryonic and environmental sources enter one response functional; a cut label cannot select the governing law.

2 Three Claim Layers

The first layer is the finite-source theorem surface: collar localization, modular-source matching, tensor tomography, quotient invariance, and refinement control. These results do not produce dark matter by themselves.

The second layer is the static QUMOND continuation. It gives the empirical RAR shape and BTFR asymptote under explicit premises, is non-held-out and exactly degenerate in (a_0, λ) , and fails Cassini when applied universally.

The third layer is the OPH-R1 completion contract. A physical parent has to cover galaxies, Solar-System tests, wide binaries, lensing, mergers, FLRW evolution, and perturbations with one source definition. Its emission from OPH microphysics and its joint likelihood are work in progress.

3 Finite Covariant Parent Contract

The cosmology layer needs more than scalar rows. The object consumed by a Boltzmann solver must be a finite covariant parent from which stress, exchange, response, and perturbation variables are calculated. At regulator r , write

$$\mathfrak{P}_r = (\mathcal{X}_r, g_r, e_r, \Pi_r, Z_r, f_r, \mathsf{T}_r, \mathsf{C}_r, \mathsf{M}_r, \mathsf{R}_r).$$

Here $\mathcal{X}_r$ is a finite spacetime or cell complex, g_r the regulated metric, e_r tetrads, Π_r parallel transport between adjacent cells, $Z_r = Z_{A,r} \sqcup Z_{R,r}$ anomaly and recipient packet states, $f_{I,z} \geq 0$ packet occupation, T_r finite covariant transport, C_r reaction or repair, M_r the moment map to currents and stress, and R_r the reconstruction/refinement comparison map. A packet $z = (I, x, p, \lambda)$ must satisfy

$$g_{\mu\nu}(x)p^\mu p^\nu = -m_I^2, \quad p^0 > 0,$$

with declared finite invariant phase-space weights w_z . For anomaly packets the Standard Model currents vanish:

$$q_z^{\text{EM}} = q_z^{\text{color}} = q_z^{\text{weak}} = 0,$$

or a separate neutral-commutant theorem must prove that the anomaly algebra induces no Standard Model current.

The finite moments are

$$J_{I,r}^\mu(x) = \frac{1}{V_x} \sum_{z: x_z=x} w_z f_{I,z} p_z^\mu, \quad T_{I,r}^{\mu\nu}(x) = \frac{1}{V_x} \sum_{z: x_z=x} w_z f_{I,z} p_z^\mu p_z^\nu.$$

If the anomaly stress cannot be represented by positive on-shell packets, the parent must declare an internal packet stress $\Sigma_z^{\mu\nu}$ and replace $p^\mu p^\nu$ by $p^\mu p^\nu + \Sigma_z^{\mu\nu}$. Thus the paper allows three parent branches: positive kinetic parent, kinetic parent with internal stress, or covariant action parent. It does not assume every desired effective stress is pressureless positive dust.

On the positive kinetic branch, $w_z f_{I,z} \geq 0$ and future causal momenta imply $T_{ab} t^a t^b \geq 0$ for every future timelike t^a . This is the packet-parent source of weak-energy positivity; it must not be manufactured by taking absolute values of scalar source rows.

Finite total-stress closure. Assume finite covariant summation by parts, equal-and-opposite face fluxes after parallel transport, reaction-channel four-momentum conservation

$$\sum_{I,z} \nu_{\alpha, I, z} p_z^\mu = 0,$$

and moment-map compatibility with the discrete connection. Multiplying the finite kinetic equation by $w_z p_z^\nu$ and summing gives

$$\text{Div}_r T_{I,r}^{\mu\nu} = Q_{I,r}^\nu, \quad Q_{I,r}^\nu(x) = \frac{1}{V_x} \sum_{z: x_z=x} w_z p_z^\nu C_{I,r,z}[f].$$

Summing over sectors cancels all reaction channels, so

$$\text{Div}_r \sum_I T_{I,r}^{\mu\nu} = 0.$$

Continuum promotion additionally requires

$$\|\mathbf{R}_r(\text{Div}_r T_r) - \nabla_\mu \mathbf{R}_r(T_r)^{\mu\nu}\|_{\mathcal{W}} \leq \epsilon_{\text{div}}(r), \quad \epsilon_{\text{div}}(r) \rightarrow 0.$$

FLRW reduction and exchange. If the homogeneous packet distribution is invariant under a spatial group with no invariant vector and no invariant traceless symmetric rank-two tensor, then

$$\bar{q}_A^\mu = 0, \quad \bar{\pi}_A^{\mu\nu} = 0, \quad \bar{T}_A^\mu{}_\nu = \text{diag}(-\bar{\rho}_A, \bar{p}_A, \bar{p}_A, \bar{p}_A).$$

Reserve Q_A^μ for exchange four-current:

$$Q_A^\mu = Q_A u^\mu + F_A^\mu, \quad u_\mu F_A^\mu = 0, \quad \sum_I Q_I^\mu = 0.$$

The conserved homogeneous anomaly load should be denoted separately, for example $\mathcal{N}_A = a^3 \bar{\rho}_A V_{\text{com}}$.

Retarded response. Linearizing the parent around the FLRW background gives

$$\delta x' = A(k, \eta) \delta x + B(k, \eta) \delta n_b^{(A)}, \quad \delta y_A = C(k, \eta) \delta x + D(k, \eta) \delta n_b^{(A)},$$

where $n_b^{(A)} = -u_{A\mu} J_b^\mu$. With $\partial_\eta U(\eta, \eta') = A(\eta) U(\eta, \eta')$,

$$K_A(\eta, \eta') = C(\eta) U(\eta, \eta') B(\eta') + D(\eta) \delta(\eta - \eta')$$

is the actual retarded response kernel. A scalar $B_A(k, a)$ and a scalar $\Gamma_{\text{rec}}(k, a)$ are controlled reductions of this realization, not primitive inputs. If there exists $H > 0$ and $\gamma > 0$ with

$$HA + A^\dagger H \preceq -2\gamma H,$$

then $\|U(\eta, \eta')\|_H \leq e^{-\gamma(\eta - \eta')}$. A one-pole relaxation rate is valid only when $\|K_A - K_A^{(1)}\|_{\mathcal{W}} \leq \epsilon_{\text{one pole}}$ in a predeclared norm.

Abundance selector boundary. The conservation equation

$$\bar{\rho}'_A + 3\mathcal{H}\bar{\rho}_A = 0$$

implies $\bar{\rho}_A = C_A a^{-3}$, but it does not determine C_A . The source-only OPH abundance requires the finite selector

$$\mathcal{E}_{A,r} = F_{A,r}(\text{quotient ensemble}, P_\star, N_{\text{CRC}}, \text{release state}, \mathfrak{P}_r), \quad \bar{\rho}_{A,r}(a) = \frac{\mathcal{E}_{A,r}}{a^3 V_{\text{com}}},$$

with refinement compatibility. Section 18.15 defines this selector. Promotion to a source-only abundance requires `ANOMALY_ABUNDANCE_SOURCE_RECEIPT`; without that receipt, cosmological abundance is a conditional source state rather than a source-only OPH prediction.

Receipt	Claim label
PACKET_MASS_SHELL_RECEIPT	finite-parent evidence gate
TOTAL_STRESS_CLOSURE_RECEIPT	finite-parent evidence gate
RETARDED_RESPONSE_RECEIPT	response-realization gate
B_A_KERNEL_RECEIPT	conditional on response and mode calibration
ANOMALY_ABUNDANCE_SOURCE_RECEIPT	source-side finite selector gate

Canonical finite covariant collar-packet parent. For a regulator r and background (X, g) , the source object is

$$\mathcal{P}_r[X, g] = (\mathcal{C}_r, g_r, e_r, U_r, Z_r, \omega_r, p_r, f_r, \Phi_r, \mathcal{R}_r, \mathcal{G}_r, \pi_r, \text{Read}_r).$$

Here $\mathcal{C}_r$ is a finite causal cell complex with oriented faces, cell four-volumes, causal adjacency, and finite parallel transports $U_{F \rightarrow c}$; e_r is the tetrad/local-frame data; $Z_r = \bigsqcup_s Z_{s,r}$ is the packet state space for anomaly, recipient, radiation, standard matter, and any interaction sector; ω_r, p_r, f_r are invariant weights, local momenta, and occupations; Φ_r is the face flux; $\mathcal{R}_r$ is the reaction channel list; $\mathcal{G}_r$ is the finite gauge/quotient data; π_r is the restriction/refinement map; and Read_r is the local observable readout. With signature $(-+++)$, every packet used in a kinetic stress branch must satisfy $g_{ab}p_z^a p_z^b = -m_z^2$, $p_z^0 > 0$, and $\omega_z f_{c,z} \geq 0$, with the invariant measure convention declared explicitly.

Scalar-to-stress nonuniqueness. A scalar collar row, even a finite row for $\rho_A(a)$, $\rho_{A,\text{eq}}(a)$, or $B_A(k, a)$, does not determine a stress tensor. Many inequivalent packet parents can have the same scalar density while differing in pressure, sound speed, anisotropic stress, exchange current, gauge-invariant perturbation variables, and causal response. Therefore no evidence bundle may promote scalar rows to physical CMB, lensing, growth, or S_8 inputs unless the parent supplies the missing stress and response variables.

Finite MaxEnt lift. When only finitely many one-cell source readings $e_x(u_i) = T_{ab}^A(x)u_i^a u_i^b$ are available, a stress lift is not unique until the branch supplies a finite selection rule. The admissible MaxEnt lift minimizes no hidden observational residual: it maximizes the declared packet entropy subject to the source readings, mass shells, conserved charges, local frame covariance, and refinement restrictions. The lift is physical only when held-out timelike probes pass a quadraticity/tomography test and the variational stress agrees with the packet-moment stress in the predeclared norm.

Packet stress and exchange closure. For a kinetic parent,

$$T_{I,r}^{ab}(c) = \frac{1}{V_c} \sum_{z \in Z_{I,r}(c)} \omega_z f_{c,z} p_z^a p_z^b, \quad J_{I,r}^a(c) = \frac{1}{V_c} \sum_{z \in Z_{I,r}(c)} \omega_z f_{c,z} p_z^a.$$

More general packet parents may add an internal stress Σ_z^{ab} , but then Σ_z^{ab} is part of the primitive parent data and must obey the same local-frame and refinement checks. The finite transport/reaction equation must imply

$$\text{Div}_r T_{I,r}^{ab} = Q_{I,r}^b, \quad \sum_I Q_{I,r}^b = 0.$$

If the repair exchange is nonzero, a row labelled “recipient” is not enough: the recipient stress T_R^{ab} , exchange current Q_R^b , and closure residual $Q_A^b + Q_R^b$ must be emitted and certified. If exchange is off, the parent instead emits a repair-exchange-off receipt.

Response, rate, and gauge receipts. The finite causal response must provide a finite domain of dependence, subluminal characteristic bound, retarded support residual, and response stability residual. The transition-matrix diagnostic may be called $\gamma_{\text{repair step}}$ until the active fiber, conserved-sector decomposition, physical clock $\Delta\tau_{\text{physical}}$, and common parent response pole are certified. Only then may one write

$$\Gamma_{\text{rec}} = -\text{Re } \lambda_{\text{active}} / \Delta\tau_{\text{physical}}.$$

Gauge checks must be stated in gauge-invariant or rest-frame variables; raw Newtonian/synchronous agreement is a diagnostic, not a gauge-independence receipt.

Receipt hierarchy. The finite parent receipt is the conjunction of primitive evidence receipts, not a producer-declared boolean. The source-side conjunction includes

SOURCE_ROUTE, ENTROPY_UNIT, MODULAR_NONADDITIVITY_IDENTITY,
SOURCE_LOCALIZATION, STRESS_TOMOGRAPHY, FINITE_PACKET_KINEMATICS,
PACKET_MASS_SHELL, TRANSPORT_COVARIANCE, CHANNEL_FOUR_MOMENTUM,
FINITE_PACKET_STRESS_READOUT, VARIATIONAL_MOMENT_STRESS_AGREEMENT,
LOCAL_FRAME_COVARIANCE, CARRIER_QUOTIENT_INVARIANCE, TOTAL_STRESS_CLOSURE,
EXCHANGE_CURRENT_CLOSURE, COSMOLOGICAL_GAUGE_INVARIANCE,
FINITE_DOMAIN_OF_DEPENDENCE, SUBLUMINAL_CHARACTERISTICS,
RETARDED_RESPONSE, RESPONSE_STABILITY, REFINEMENT_CONVERGENCE, CDM_LIMIT_RECOVERY.

Frozen source, solver, likelihood, and data hashes are later promotion receipts. They do not belong to the parent theorem and cannot rescue a failed parent.

4 Source Ontology, Units, and Matching

This paper uses metric signature $(-+++)$. Entropies are in nats unless a receipt explicitly declares bits and multiplies by $\ln 2$. The radius ℓ in the local source formula is the proper geodesic radius of a small four-dimensional causal diamond or ball read in the local tetrad, with scale hierarchy

$$\ell_{\text{UV}} \ll \delta_{\text{collar}} \ll \ell \ll L_{\text{curvature}}, L_{\text{gradient}}.$$

Natural units may set $\hbar = c = 1$. In SI units a dimensionless entropy source R has energy-density normalization

$$\rho_A c^2 = \frac{15\hbar c}{8\pi^2 \ell^4} R.$$

4.1 Three Source Objects

The dark continuation distinguishes

$$R_C^{\text{info}} := I_\rho(A : D|B), \quad \mathcal{M}_C^{\text{anom}}, \quad T_A^{ab}.$$

The first is raw finite conditional mutual information. The second is a source variable proved, or explicitly hypothesized on the continuation branch, to equal the anomalous local modular energy entering the Bisognano-Wichmann (BW) ball calculation. The third is a tensor stress reconstructed from all-frame source readings and a finite packet parent. The paper does not identify these three objects by notation alone.

Exact finite modular-nonadditivity identity. For a faithful finite-dimensional state ρ_{ABD} , define support modular Hamiltonians $K_X = -\log \rho_X$, embedded in the full algebra, and

$$\Delta K_\rho = K_{AB} + K_{BD} - K_B - K_{ABD}.$$

Then

$$\text{Tr } \rho_{ABD} \Delta K_\rho = S(AB) + S(BD) - S(B) - S(ABD) = I_\rho(A : D|B).$$

This is an expectation identity. It does not make ΔK_ρ a positive operator, a state-independent geometric generator, or a local collar stress.

First-variation obstruction. If σ is full rank with $I_\sigma(A : D|B) = 0$ and $\rho(t) = \sigma + tX + O(t^2)$, then

$$\left. \frac{d}{dt} I_{\rho(t)}(A : D|B) \right|_{t=0} = 0$$

because the nonnegative CMI functional has a local minimum on the exact Markov set. A fixed-reference modular-energy variation $\delta \langle K_\sigma \rangle = t \text{Tr}(X K_\sigma) + O(t^2)$ is generically linear. Therefore raw CMI is not generically the same object as the anomalous local modular energy on a perturbative Markov branch.

4.2 Declared Source Route

A source artifact must declare exactly one route:

$$\text{source_route} \in \{\text{FIXED_REFERENCE_MODULAR_ENERGY}, \text{NONLINEAR_CMI_STRESS}\}.$$

There is no automatic mixed mode. On the fixed-reference route the source is

$$R_C^{\text{src}} = \mathcal{M}_C^{\text{anom}} = \delta \langle K_C^{(\text{anom})}[\sigma] \rangle,$$

and CMI is a recovery/localization diagnostic. On the nonlinear-CMI route the source is $R_C^{\text{src}} = R_C^{\text{info}}$ only after a named CMI-to-modular-source matching theorem or explicit continuation hypothesis with residual ledger.

Target 4.1 (CMI-to-modular-source matching). *The nonlinear-CMI route must prove or declare a versioned hypothesis that, for each small ball $B_\ell(x, u)$,*

$$R_{x,u,\ell}^{\text{info}} = \frac{2\pi}{\hbar c} \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell} T_{ab}^A u^a u^b dV + \varepsilon_{\text{match}},$$

with refinement-stable matching residual. Strong subadditivity alone does not prove this statement. With a subtracted background value, positivity requires a separate packet-parent or branch-restriction certificate.

Theorem 4.2 (Conditional local BW source normalization). *Assume the scale hierarchy above, a declared source route, the matching/local modular representation*

$$R_{x,u,\ell}^{\text{src}} = \frac{2\pi}{\hbar c} \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell} T_{ab}^A u^a u^b dV + \varepsilon_{\text{match}},$$

and smooth $\varepsilon_A(x, u) = T_{ab}^A(x)u^a u^b$ on the ball. Then

$$R_{x,u,\ell}^{\text{src}} = \frac{8\pi^2 \ell^4}{15\hbar c} \left[\varepsilon_A(x, u) + \frac{\ell^2}{14} D_{\perp}^2 \varepsilon_A(x, u) + O\left(\frac{\ell^4}{L_{\text{gradient}}^4} + \frac{\ell^2}{L_{\text{curvature}}^2} \right) \right] + \varepsilon_{\text{match}}.$$

Hence, on the controlled local branch,

$$\varepsilon_A(x, u) = \frac{15\hbar c}{8\pi^2} \lim_{\ell^4} \frac{R_{x,u,\ell}^{\text{src}}}{\ell^4}$$

with the stated finite-ball, curvature, and matching corrections.

Proof. The BW ball weight is $w_{\ell}(r) = (\ell^2 - r^2)/(2\ell)$. In flat Riemann-normal coordinates

$$\int_{B_{\ell}} w_{\ell}(r) d^3x = 4\pi \int_0^{\ell} r^2 \frac{\ell^2 - r^2}{2\ell} dr = \frac{4\pi \ell^4}{15}.$$

Multiplying by $2\pi/(\hbar c)$ gives $8\pi^2 \ell^4/(15\hbar c)$. The next term follows from angular averaging $\varepsilon_A(y) = \varepsilon_A(x) + r^2 D_{\perp}^2 \varepsilon_A(x)/6 + \dots$ and $\int_{B_{\ell}} w_{\ell}(r) r^2 d^3x = 4\pi \ell^6/35$, whose ratio to the constant term gives $\ell^2/14$. Curvature corrections are the usual Riemann-normal-coordinate corrections to the volume form and kernel. $\square$

4.3 Localization, Covers, and Tomography

Let $E_{B,\text{phys}}$ be the conditional expectation onto the quotient-visible physical collar algebra. A finite source receipt must bound

$$|R_C^{\text{src}} - \langle E_{B,\text{phys}}(\Delta K) \rangle| \leq \varepsilon_{\text{loc}}$$

with hidden-carrier, port-relabeling, repair-schedule, and refinement invariance. Cover independence is a separate requirement: admissible collar covers must converge to the same weak source measure, or a Möbius/inclusion-exclusion valuation must prove that overlapping collars do not double count the same information defect.

A scalar reading for one observer gives only $T_A^{ab}u_a u_b$. To reconstruct a symmetric tensor, the source must be read on enough timelike directions. For $e_x(u) = \lim(15\hbar c/8\pi^2)R_{x,u,\ell}^{\text{src}}/\ell^4$, define the homogeneous extension

$$Q_x(v) = [-g(v, v)]e_x\left(v/\sqrt{-g(v, v)}\right).$$

If Q_x extends quadratically and satisfies the parallelogram identity, there is a unique symmetric T_{ab}^A recovered by polarization. The finite simulator version needs at least ten linearly independent timelike probes and held-out quadraticity tests; a single $u^{\mu} = (1, 0, 0, 0)$ row is not a stress theorem.

5 The Dark Matter Challenge

In Newtonian gravity and in the weak-field limit of general relativity, a test particle in a circular orbit satisfies

$$\frac{v^2(r)}{r} = g(r) = \frac{GM(< r)}{r^2}. \quad (1)$$

If almost all gravitating matter is visible baryonic matter, then outside the bright disk the enclosed mass should stop growing quickly and the rotation speed should fall as $v(r) \propto r^{-1/2}$. Disk galaxies

do not behave that way. Their outer rotation curves are often flat, so the inferred dynamical mass grows approximately linearly with radius:

$$M_{\text{dyn}}(< r) = \frac{v^2 r}{G}. \quad (2)$$

The galaxy problem is only one part of the evidence. Clusters need more gravity than the observed gas and galaxies provide. Weak lensing maps in merging systems such as the Bullet Cluster show gravitational potential peaks separated from the X-ray gas [19]. The CMB and large-scale structure prefer a gravitating component that is mostly non-baryonic, close to pressureless, and present before galaxies settle [5]. Planck’s base Λ CDM parameter set gives $\Omega_c h^2 = 0.120 \pm 0.001$ and $\Omega_b h^2 = 0.0224 \pm 0.0001$, so most matter is not baryonic in that model [5].

The challenge includes more than “flat rotation curves”. A good theory has to explain:

- galaxy rotation curves and the baryonic Tully-Fisher relation;
- the tight radial acceleration relation, abbreviated RAR;
- galaxy and cluster lensing;
- cluster offsets in mergers;
- CMB acoustic peaks, BAO, matter power, weak lensing, and S_8 .

6 Standard Routes

6.1 Cold Dark Matter

The standard route adds an additional non-baryonic matter component. In its simplest form it is cold, effectively collisionless, and pressureless on cosmological scales. Λ CDM uses the same component for CMB peaks, large-scale structure, clusters, lensing, and galaxy halos.

The particle identity is absent. Direct detection has produced no confirmed dark matter particle. Galaxy data show a tight connection between baryonic acceleration and observed acceleration, much tighter than a naive halo-plus-disk picture suggests. Halo fitting also introduces galaxy-level nuisance freedom, while the RAR looks like a law.

6.2 MOND

MOND should be treated as a clue, not as a failure. Its basic phenomenology starts from a critical acceleration a_0 , and in the deep low-acceleration regime it gives

$$g \simeq \sqrt{a_0 g_b}. \quad (3)$$

For a point baryonic mass this implies

$$v^4 = G M_b a_0, \quad (4)$$

which is exactly the baryonic Tully-Fisher scaling [14]. This is MOND’s great success: it goes straight at the galaxy regularity.

The pressure points start when the same static acceleration rule is asked to do more jobs than a galaxy equilibrium law can naturally do:

- *Relativistic completion and lensing.* A nonrelativistic force law does not by itself say which metric photons follow, or how the modified field contributes to the stress side of Einstein's equations. Relativistic completions such as TeVeS exist [15], but the lensing sector then depends on extra fields and closure assumptions.
- *Clusters.* MOND reduces the cluster mass discrepancy but does not remove the need for additional gravitating stress in clusters. Merging systems are especially sharp: if the anomaly is an instantaneous function of the local baryonic acceleration, the lensing peak is too tightly tied to the visible baryonic source.
- *The early universe.* The CMB acoustic peaks, BAO, matter growth, CMB lensing, and weak-lensing amplitudes need a homogeneous and perturbative gravitating component before settled galaxies exist. A static RAR law does not by itself define that background abundance or its linear response.
- *Non-spherical and time-dependent systems.* The algebraic relation $g \simeq \sqrt{a_0 g_b}$ is clean in high-symmetry systems. In real three-dimensional fields, curl terms, environmental dependence, transport, and relaxation can matter.

The implemented OPH galaxy law is QUMOND, not a derived stress sector. It does not determine relativistic slip, transported cluster stress, a homogeneous abundance, or cosmological perturbations. The MOND pressure points therefore are physical requirements. OPH-R1 states one response contract under which all of them have to be calculated from the same finite self-reading source and kernel.

7 The OPH Route

Observer-Patch Holography supplies finite collars, central record registers, local recovery maps, modular readout, quotient observables, and a recovered Einstein branch. The dark continuation asks whether a repair remainder can generate a physical gravitational response. Its logical chain is

finite collar diagnostic $\longrightarrow$ localized modular source $\longrightarrow$ covariant response action $\longrightarrow$ metric and stress observables

Raw CMI is not a stress tensor, a settled normal form is not a probability law, and a QUMOND virtual density is not automatically a positive matter density.

Proposition A, QUMOND identity. The static equation below is Milgrom's QUMOND equation. Its action and conservation properties are imported mathematics. OPH has to derive the interpolation, scale, applicability, and covariant parent from its source.

Proposition B, exact static non-identifiability. For

$$\nu_{a_0, \lambda}(g) = \frac{1}{1 - \exp[-\lambda \sqrt{g/a_0}]},$$

the identity

$$\nu_{a_0, \lambda}(g) = \nu_{a_0/\lambda^2, 1}(g)$$

holds for every $g \geq 0$. Static observables depend on a_0 and λ only through $a_{0, \text{eff}} = a_0/\lambda^2$.

Proposition C, signed-source fork. The QUMOND virtual density is positive for the isolated point-source solution below but can be negative in nonspherical, multi-source, or external-field configurations. It cannot generally equal a positive on-shell packet density.

Proposition D, theory status. The universal static continuation is excluded by Cassini, and the closed-cut predicate is invalid. OPH-R1 is a completion contract whose microphysical kernel, relativistic action, and likelihood implementation are work in progress.

8 Static Galaxy Derivation

8.1 Source From The Carried Collar Remainder

On the support-visible BW branch the primary theorem is the cap automorphism identity. In the special type-I or effective local representation where the geometric generator may be written inside the cap algebra, the OPH modular decomposition has the local form

$$K_C = 2\pi B_C + K_C^{(\text{anom})} + \text{const.} \quad (5)$$

At finite regulator stage, the nonadditive modular part is carried on an overlap collar. The dark branch may use the raw finite diagnostic

$$R_C^{\text{info}} = I_\omega(A : D|B) \geq 0,$$

but the physical density formula uses the routed source variable R_C^{src} . On the fixed-reference route this is the anomalous modular-energy variation. On the nonlinear-CMI route it is raw CMI only after the explicit CMI-to-modular-source matching theorem or hypothesis and its localization, cover, and tomography receipts pass. With those receipts, the effective rest-energy density is written

$$\rho_{Ac}^2 = \frac{15\hbar c}{8\pi^2 \ell^4} R_C^{\text{src}}, \quad \rho_A = \frac{15\hbar}{8\pi^2 c \ell^4} R_C^{\text{src}}. \quad (6)$$

Here ℓ is the proper geodesic radius of the local four-dimensional small ball. It is not a cap angle, a screen spacing, a collar thickness, a coordinate radius, or a fitted galaxy scale. If the simulator records CMI in bits, the value must be multiplied by $\ln 2$ before using (6). The weak-field Poisson equation becomes

$$\nabla^2 \Phi = 4\pi G(\rho_b + \rho_A). \quad (7)$$

Conditional theorem 1, attractive sign. If the routed source R_C^{src} is nonnegative on the declared branch, then the anomaly is attractive in the Newtonian weak-field equation. For the nonlinear-CMI route this nonnegativity is inherited from raw CMI only in the absence of background subtraction; otherwise it must be certified by the packet parent or branch restriction.

Proof. The premise and (6) give $\rho_A \geq 0$. Equation (7) then shows that the anomaly enters with the same attractive sign as baryonic mass. $\square$

Finite-source receipt boundary. The exact CMI modular identity, first-variation obstruction, declared route, matching theorem or hypothesis, collar localization, BW normalization, ℓ^4 plateau, cover independence, and timelike tomography receipts are the source-side content of Issue #319. Without them, R_C^{info} is a recoverability/readback diagnostic and not a physical density source. After them, a finite positive packet parent must supply T_A^{ab} , its Standard-Model-neutral charge condition, and the conserved anomaly plus recipient stress closure.

8.2 The Acceleration Variable

Define the baryonic acceleration ratio

$$x := \frac{g_b}{a_{0,\text{OPH}}}. \quad (8)$$

For a point baryonic source,

$$g_b(r) = \frac{GM_b}{r^2}, \quad r_M := \sqrt{\frac{GM_b}{a_{0,\text{OPH}}}}, \quad (9)$$

so

$$x = \left(\frac{r_M}{r}\right)^2, \quad \sqrt{x} = \frac{r_M}{r}. \quad (10)$$

The OPH support assumption is that settled-galaxy repair opportunities live on codimension-one collar/cut support. A screen-area count would scale as x . A cut-count scales as $\sqrt{x}$, as in (10).

8.3 Poisson Activation

Let the mean number of independent local repair opportunities be

$$\mu(x) = \lambda_{\text{collar}} \sqrt{x}. \quad (11)$$

The Poisson law does not follow from fixing this mean alone. A fixed mean selects no distribution. Maximizing Shannon entropy over nonnegative integers at fixed mean gives a geometric distribution, not a Poisson distribution. The OPH claim therefore needs independent increments on disjoint collar intervals, rare local repair events, and refinement stability. The following model states those premises and carries an explicit finite-depth total-variation bound.

Definition 8.1 (Refinement-stable collar opportunity model). *Fix a cut interval I and a finite intensity measure μ on I . For each refinement depth $r \in \mathbb{N}$, partition I into finitely many collar cells $B_{r,1}, \dots, B_{r,m_r}$, and attach to each cell an activation indicator $X_{r,i} \in \{0, 1\}$ with $p_{r,i} := \Pr[X_{r,i} = 1]$. For every union of cells $J \subseteq I$ write*

$$N_r(J) := \sum_{B_{r,i} \subseteq J} X_{r,i}, \quad \mu_r(J) := \sum_{B_{r,i} \subseteq J} p_{r,i}.$$

The family is a refinement-stable collar opportunity model when three premises hold.

1. *Disjoint-collar independence: at every depth r the indicators $X_{r,1}, \dots, X_{r,m_r}$ are independent.*
2. *Rare local activation: $p_{\max}(r) := \max_i p_{r,i} \rightarrow 0$ as $r \rightarrow \infty$.*
3. *Refinement stability: there is a sequence $\delta_r \rightarrow 0$ with $|\mu_r(J) - \mu(J)| \leq \delta_r$ for every cell-aligned $J \subseteq I$ and every depth r .*

Each cell carries an indicator, not a count, so multiplicity inside a cell is resolved by passing to deeper cells; refinement stability is exactly the statement that this re-counting reproduces the same intensity measure at every depth.

Theorem 8.2 (Quantitative Poisson law for collar opportunities). *In a refinement-stable collar opportunity model, for every cell-aligned subinterval $J \subseteq I$ and every depth r ,*

$$d_{\text{TV}}(\mathcal{L}(N_r(J)), \text{Poisson}(\mu(J))) \leq \sum_{B_{r,i} \subseteq J} p_{r,i}^2 + \delta_r \leq \mu_r(J) p_{\max}(r) + \delta_r, \quad (12)$$

with $d_{\text{TV}}(\alpha, \beta) := \sup_{A \subseteq \mathbb{N}} |\alpha(A) - \beta(A)|$. *The right side vanishes as $r \rightarrow \infty$, so $N_r(J)$ converges in total variation to $\text{Poisson}(\mu(J))$: for every n ,*

$$\lim_{r \rightarrow \infty} \Pr[N_r(J) = n] = e^{-\mu(J)} \frac{\mu(J)^n}{n!}. \quad (13)$$

Proof. Le Cam's inequality [16, 17] bounds the total-variation distance between a sum of independent Bernoulli indicators and the Poisson law with the matching mean:

$$d_{\text{TV}}(\mathcal{L}(N_r(J)), \text{Poisson}(\mu_r(J))) \leq \sum_{B_{r,i} \subseteq J} p_{r,i}^2.$$

Disjoint-collar independence supplies the independence premise. The monotone coupling of two Poisson laws gives $d_{\text{TV}}(\text{Poisson}(a), \text{Poisson}(b)) \leq |a - b|$, and refinement stability bounds $|\mu_r(J) - \mu(J)|$ by δ_r . The triangle inequality gives the first inequality in (12); the second uses $p_{r,i}^2 \leq p_{\max}(r) p_{r,i}$. Rare local activation sends $\mu_r(J) p_{\max}(r)$ to zero because $\mu_r(J) \leq \mu(J) + \delta_r$ stays bounded, and $\delta_r \rightarrow 0$ by refinement stability. $\square$

Corollary 8.3 (Poisson process limit and λ_{collar} identification). *In a refinement-stable collar opportunity model, for disjoint cell-aligned subintervals $J_1, \dots, J_k \subseteq I$,*

$$d_{\text{TV}}(\mathcal{L}(N_r(J_1), \dots, N_r(J_k)), \bigotimes_{s=1}^k \text{Poisson}(\mu(J_s))) \leq \sum_{s=1}^k [\mu_r(J_s) p_{\max}(r) + \delta_r] \rightarrow 0,$$

so the limiting counting process on I is the Poisson process with intensity measure μ . On codimension-one collar/cut support the total intensity at acceleration ratio x is $\mu(x) = \lambda_{\text{collar}} \sqrt{x}$, and λ_{collar} is identified as the refinement-stable active-opportunity intensity per unit of the dimensionless cut coordinate $\sqrt{x} = r_M/r$. The numerical value of λ_{collar} is branch data from the finite-collar coefficient section, not an output of this corollary.

Proof. The count vectors on disjoint cell-aligned subintervals are independent by disjoint-collar independence, and total variation is subadditive over products of independent coordinates, so the joint distance is bounded by the sum of the marginal bounds (12). Kolmogorov consistency of the limiting finite-dimensional laws defines the Poisson process with intensity μ . The identification of λ_{collar} restates the codimension-one cut count of (10) with its refinement-stable proportionality constant. $\square$

In particular, the zero-count probability in the refinement limit is

$$\Pr[N = 0] = e^{-\mu(x)} = \exp[-\lambda_{\text{collar}} \sqrt{x}], \quad (14)$$

and the probability that at least one repair opportunity is active is

$$p(x) = 1 - \Pr[N = 0] = 1 - \exp[-\lambda_{\text{collar}} \sqrt{x}]. \quad (15)$$

At finite depth, (12) bounds the activation-probability error itself: $|\Pr[N_r(I) \geq 1] - p(x)| \leq \mu_r(I) p_{\max}(r) + \delta_r$.

Finite-range dependence robustness. Disjoint-collar independence can be weakened without losing the limit. Suppose each $X_{r,i}$ is independent of $\sigma(X_{r,j} : j \notin B_i)$ for a dependence neighborhood $B_i \subseteq \{1, \dots, m_r\}$ with $|B_i| \leq 2R_r + 1$ (finite dependence range R_r in cells). The Chen–Stein bound of Arratia, Goldstein, and Gordon [18] gives

$$d_{\text{TV}}(\mathcal{L}(N_r(J)), \text{Poisson}(\mu_r(J))) \leq b_1 + b_2, \quad b_1 = \sum_i \sum_{j \in B_i} p_{r,i} p_{r,j}, \quad b_2 = \sum_i \sum_{j \in B_i \setminus \{i\}} \mathbb{E}[X_{r,i} X_{r,j}],$$

with the mixing term $b_3 = 0$ under exact neighborhood independence. With $b_1 \leq (2R_r + 1) p_{\max}(r) \mu_r(J)$ and $b_2 \leq 2R_r \chi_r \mu_r(J)$, where $\chi_r := \max_i \max_{j \in B_i \setminus \{i\}} \Pr[X_{r,j} = 1 \mid X_{r,i} = 1]$, the Poisson limit holds whenever $R_r p_{\max}(r) \rightarrow 0$ and $\chi_r \rightarrow 0$. A finite-range Gibbs collar model with exponential clustering sits in this class; the collar CMI decay packet is the corresponding recoverability statement.

Machine witness. The certificate `collar_poisson_witness_certificate.json` (code/dark_matter/certifi generator and verifier alongside) enumerates exact opportunity-count laws on declared finite families: uniform and nonuniform independent families at the unit-branch and $\mathbb{Z}_6$ -presence-branch intensities, a mean-drift family exercising the δ_r term, and a one-dependent family exercising the $b_1 + b_2$ route. For each family it reports the exact total-variation distance to the matching Poisson law and checks it against (12) or $b_1 + b_2$.

The activation count is only the counting part of the static law. The source closure says that active scalar repair opportunities recover baryonic flux in the settled one-dimensional branch:

$$g_b = p(x) g_{\text{obs}}. \quad (16)$$

Thus

$$g_{\text{obs}} = \nu_{\text{OPH}}(x) g_b, \quad \nu_{\text{OPH}}(x) = \frac{1}{1 - \exp[-\lambda_{\text{collar}} \sqrt{x}]}. \quad (17)$$

At $\lambda_{\text{collar}} = 1$ this is the McGaugh–Lelli–Schombert empirical interpolating function fitted to SPARC [8]; SPARC comparisons of (17) are therefore non-held-out. The universal extrapolation of this static law to a Solar source in the Galactic external field is excluded by Cassini at 19.22σ (18.01σ on the unit branch; see the Cassini Applicability Stress subsection). The equation is a galaxy-domain empirical diagnostic; no source-derived scope theorem makes it a physical galaxy-only law.

Conditional theorem 2, activation law. Assume codimension-one collar/cut support, a refinement-stable collar opportunity model in the sense of Definition 8.1 on the settled-galaxy cut, and the settled scalar flux-recovery closure (16). Then the settled one-dimensional galaxy response is (17).

Proof. Codimension-one support gives a mean count proportional to $\sqrt{x}$. The scalar coefficient is λ_{collar} . Theorem 8.2 and Corollary 8.3 give the Poisson law (13) with mean $\mu(x) = \lambda_{\text{collar}} \sqrt{x}$. The active probability is the complement of the zero-count event, which is (15). Solving $g_b = p(x) g_{\text{obs}}$ in one-dimensional symmetry gives (17). $\square$

The counting step is unconditional on the declared model class: Theorem 8.2 is a theorem, not an assumption. The conditional content of conditional theorem 2 is physical: that settled-galaxy repair opportunities realize a refinement-stable collar opportunity model on codimension-one cut support, and that the flux-recovery closure (16) holds. The simulator static-galaxy proxy and the SPARC comparison scripts consume the activation law (15); under Definition 8.1 the counting step contributes no freedom to those comparisons beyond the single coefficient λ_{collar} , so a held-out SPARC test constrains λ_{collar} and the physical premises, never the Poisson shape separately.

8.4 Conservative Potential Equation

For general disks and three-dimensional baryonic profiles, the algebraic vector rule

$$\mathbf{g}_{\text{alg}} = \nu_{\text{OPH}}(|\mathbf{g}_b|/a_{0,\text{OPH}})\mathbf{g}_b$$

need not be curl-free because

$$\nabla \times \mathbf{g}_{\text{alg}} = \nabla \nu_{\text{OPH}} \times \mathbf{g}_b.$$

The conservative equation used here is

$$\nabla^2 \Phi = \nabla \cdot \left[\nu_{\text{OPH}} \left(\frac{|\nabla \Phi_b|}{a_{0,\text{OPH}}} \right) \nabla \Phi_b \right], \quad \nabla^2 \Phi_b = 4\pi G \rho_b, \quad \mathbf{g}_{\text{obs}} = -\nabla \Phi. \quad (18)$$

This is QUMOND [20]. Up to the potential sign convention, it follows from the bi-potential quasi-static action

$$S_Q[\Phi, \Phi_b] = -\frac{1}{8\pi G} \int d^3x [2\nabla \Phi \cdot \nabla \Phi_b - a_0^2 \mathcal{Q}(Y)] - \int d^3x \rho_b \Phi, \quad Y = \frac{|\nabla \Phi_b|^2}{a_0^2}, \quad \mathcal{Q}'(Y) = \nu(\sqrt{Y}). \quad (19)$$

Variation with respect to Φ gives the baryonic Poisson equation, and variation with respect to Φ_b gives (18). The action and its conservation laws belong to QUMOND; they are not an OPH derivation.

Exact static non-identifiability. For every $g_b \geq 0$,

$$\lambda_{\text{collar}} \sqrt{\frac{g_b}{a_{0,\text{OPH}}}} = \sqrt{\frac{g_b}{a_{0,\text{eff}}}}, \quad a_{0,\text{eff}} = \frac{a_{0,\text{OPH}}}{\lambda_{\text{collar}}^2},$$

and therefore

$$\nu_{\text{OPH}} \left(\frac{g_b}{a_{0,\text{OPH}}}; \lambda_{\text{collar}} \right) = \frac{1}{1 - \exp[-\sqrt{g_b/a_{0,\text{eff}}}]}$$

The identity is exact. Static rotation curves, the RAR, the BTFR, and lensing calculations that reuse this potential constrain only $a_{0,\text{eff}}$, not $a_{0,\text{OPH}}$ and λ_{collar} separately.

Symmetry reduction. In a one-coordinate symmetry with flux-area factor $A(r)$,

$$\frac{1}{A(r)} \frac{d}{dr} \{A(r)[g_{\text{obs}} - \nu(g_b/a_0)g_b]\} = 0, \quad g_{\text{obs}} - \nu(g_b/a_0)g_b = \frac{C}{A(r)}.$$

The integration term is C , C/r , or C/r^2 in planar, cylindrical, or spherical symmetry. Ordinary isolated boundary and regularity conditions set $C = 0$. The SPARC tables use the pointwise algebraic diagnostic, not the full disk equation. A publication-grade comparison has to solve (18) with the same nuisance model as comparison theories. The universal equation is excluded by Cassini; system-dependent applicability switches are not part of the OPH-R1 contract.

8.5 Newtonian And Deep-IR Limits

For $x \gg 1$, $p(x) \rightarrow 1$, hence

$$g_{\text{obs}} \rightarrow g_b. \quad (20)$$

For $x \ll 1$,

$$p(x) = \lambda_{\text{collar}} \sqrt{x} + O(x), \quad (21)$$

so

$$g_{\text{obs}} \simeq \frac{g_b}{\lambda_{\text{collar}} \sqrt{g_b/a_{0,\text{OPH}}}} = \sqrt{\frac{a_{0,\text{OPH}}}{\lambda_{\text{collar}}^2}} g_b = \sqrt{a_{0,\text{eff}} g_b}, \quad (22)$$

where

$$a_{0,\text{eff}} = \frac{a_{0,\text{OPH}}}{\lambda_{\text{collar}}^2}. \quad (23)$$

For a point source,

$$v^4 = GM_b a_{0,\text{eff}}. \quad (24)$$

Theorem 3, BTFR. The OPH activation law implies the baryonic Tully-Fisher scaling in the deep-IR limit.

Proof. For a circular orbit, $v^2/r = g_{\text{obs}}$. In the deep-IR limit, $g_{\text{obs}} = \sqrt{a_{0,\text{eff}} GM_b/r^2}$. Multiplying by r gives $v^2 = \sqrt{GM_b a_{0,\text{eff}}}$, hence (24). $\square$

8.6 Effective QUMOND Source And Stress Fork

The potential equation can be rewritten with a virtual or “phantom” source:

$$\rho_{\text{ph}} = -\frac{1}{4\pi G} \nabla \cdot [(\nu_{\text{OPH}} - 1) \mathbf{g}_b], \quad \nabla^2 \Phi = 4\pi G(\rho_b + \rho_{\text{ph}}). \quad (25)$$

For a spherical system,

$$M_{\text{dyn}}(r) = \nu_{\text{OPH}}(x) M_b(r), \quad M_{\text{ph}}(r) = [\nu_{\text{OPH}}(x) - 1] M_b(r).$$

For an isolated point baryonic source,

$$M_{\text{ph}}(r) = \frac{M_b}{\exp(\lambda_{\text{collar}} r_M/r) - 1},$$

and

$$\rho_{\text{ph}}(r) = \frac{M_b \lambda_{\text{collar}} r_M \exp(\lambda_{\text{collar}} r_M/r)}{4\pi r^4 [\exp(\lambda_{\text{collar}} r_M/r) - 1]^2}.$$

These expressions are positive, but (25) can contain negative regions for nonspherical, multi-source, and external-field configurations. A signed virtual density is admissible as a rewriting of a modified potential equation; it is not a positive material fluid [22].

The covariant-action route treats ρ_{ph} as an effective metric source and obtains lensing and cosmology by varying a full action. A positive transported-stress route cannot equal ρ_{ph} pointwise in every configuration; pressure, internal stress, exchange, nonlocality, or a different metric coupling has to reproduce the same settled force. OPH-R1 uses the covariant-action route as its minimal contract and keeps positive packets as a distinct parent class.

RAR/source matching target. The local modular-source normalization and QUMOND continuation are separate claims. An OPH source theorem has to derive the quasi-static action or source functional without fitting SPARC and show that the same covariant parent produces the metric response. Equality of one scalar density row is insufficient.

9 The Finite-Collar Coefficient

9.1 Unit Branch

The default branch is

$$\lambda_{\text{collar}} = 1, \quad a_{0,\text{eff}} = a_{0,\text{OPH}}. \quad (26)$$

This branch sets the finite-collar opportunity density to one.

9.2 $\mathbb{Z}_6$ Reserve Branch

The OPH gauge structure contains the realized quotient

$$G_{\text{phys}} = \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}. \quad (27)$$

The protected shared-edge reserve has mean

$$\epsilon_{\mathbb{Z}_6} = \frac{\bar{\ell}_{\text{shared}}}{|\mathbb{Z}_6|} = \frac{P/4}{6} = \frac{P}{24}. \quad (28)$$

Using

$$P = 1.630968209403959, \quad (29)$$

gives

$$\epsilon_{\mathbb{Z}_6} = 0.06795700872516496. \quad (30)$$

Trace convention. The reserve mean is the normalized quotient expectation:

$$\tau_q(Z_{6,r,C}) = \epsilon_{\mathbb{Z}_6} = \frac{P}{24}.$$

The reciprocal reserve-depth trace is

$$\text{Tr}_q^\#(Z_{6,r,C}) = \frac{24}{P}.$$

The survival factor is the complement of the normalized mean,

$$1 - \tau_q(Z_{6,r,C}) = 1 - \frac{P}{24},$$

not a function of $\text{Tr}_q^\#(Z_{6,r,C})$.

Conditional bridge 4, quotient-edge co-registration. Galaxy scalar repair opportunities are quotient-edge cut-register events on the same edge-center resolution that carries the protected $\mathbb{Z}_6$ reserve. The $\mathbb{Z}_6$ reserve therefore co-registers with the scalar activation count.

Conditional argument. The scalar source is a quotient-local recoverability defect, so pure $\mathbb{Z}_6$ center motion changes no realized matter or Higgs state and is inactive for scalar sourcing. If the scalar count and the protected reserve are forced onto the same center-projector resolution, the reserve removes active scalar opportunities from that same layer. $\square$

Imported scalar-slot closure. This bridge is discharged, on the quotient-edge scalar branch, by the quotient-edge collar theorem stack of the screen-microphysics paper [3]. That paper defines

$$\text{Scal}_{r,C} = \mathcal{E}_{r,C}$$

on the edge-center scalar-completeness branch and proves the imported statements labeled

`thm:scalar-activation-edge-center`, `thm:scalar-z6-same-register-commute`, `thm:scalar-channel-ex`

Thus

$$\Pi_{r,C}^{\text{scal}} \in \mathcal{E}_{r,C}, \quad Z_{6,r,C} \in \mathcal{E}_{r,C}, \quad [\Pi_{r,C}^{\text{scal}}, Z_{6,r,C}] = 0,$$

and the scalar activation channel has no independent scalar bulk, screen-area, or noncentral collar carrier unless the scalar-completeness or scalar-activation-channel completeness branch fails. For a nondegenerate OPH-coherent material subfederation, the imported source readout is

$$S_{\text{coh}}^{\text{can}} = \mathbf{1}_{\text{self-read}} R_U P_U C_U > 0$$

and coherent-matter same-channel forcing places this source in $\mathcal{E}_{r,C}$. This is a scalar-channel identity theorem. It supplies no finite covariant dark-stress parent, no abundance prediction, and no Boltzmann-likelihood closure. The scalar source route is

$$R_C^{\text{src}} \mapsto \mathcal{N}_{\nu,r,C} \mapsto \nu_{\text{OPH}}.$$

The exact $\mathbb{Z}_6$ coefficient additionally requires the scalar-weighted reserve receipt supplied by the imported gate `UNIFORM_PRODUCT_THICKENING_EXACT`; co-registration alone is not the same as scalar-reserve unbiasedness.

Finite channel bridge. The same-channel statement can be represented by one finite channel object: one indexed family E carries both the record panel $(p_e(q), a_e, q \mapsto q \oplus e)$ used by the scalar opportunity count and the collar panel (w_e, ϵ_e) used by the reserve survival factor. Then the record slots and collar slices are definitionally the same finite family, and the composite law reads

$$\lambda_{\text{collar}}(\mathcal{L}^{\text{coh}} \mathcal{N}_{\nu})(q) = \left(1 - \frac{P}{24}\right) S_{\text{coh}}^{\text{can}}(q) A(q)$$

on the uniform product-thickening branch. This bridge removes the bookkeeping ambiguity in the phrase “the same counter.” It does not close the physical residues: nature must instantiate the one-channel structure, and the numerical record- ΔS to gravity- ΔS calibration remains a separate G9 obligation.

Conditional bridge 5, presence reading of local reserve survival. $Z_{6,r,C}$ is a sum of sector projections, so the conditional reserve profile $\epsilon_{\mathbb{Z}_6}(y)$, a ratio of quotient traces of projections, is the conditional probability of the reserve-presence event in a co-registered local scalar slot at transverse collar coordinate y . A scalar opportunity survives exactly when the presence event does not occur, so the local scalar survival factor is

$$\lambda_{\text{slot}}(y) = \Pr[N_{\mathbb{Z}_6}(y) = 0] = 1 - \epsilon_{\mathbb{Z}_6}(y) \tag{31}$$

for every occupancy law whose presence event has probability $\epsilon_{\mathbb{Z}_6}(y)$.

Proof. The presence event is the projection $Z_{6,r,C}$, and its conditional probability on the scalar-active slot is $\epsilon_{\mathbb{Z}_6}(y)$. Survival is the complement event. No occupancy variable with that presence probability attains the Poisson zero-count factor $\exp(-\epsilon_{\mathbb{Z}_6})$: $1 - \epsilon < e^{-\epsilon}$ strictly for $\epsilon > 0$. A mean-count reading, in which $\epsilon_{\mathbb{Z}_6}(y)$ is consumed as the expectation of an $\mathbb{N}$ -valued occupancy, severs the link to the projection trace and consumes a scalar carrier beyond $Z_{6,r,C}$, which the imported no-separate-scalar-carrier theorem excludes; its assumption-minimal consequence is the Markov band recorded in the conditional argument below. $\square$

9.3 Finite Thickness

Let y be the transverse collar coordinate and $w(y)$ the normalized activation weight:

$$\int dy w(y) = 1. \quad (32)$$

The finite-thickness branch gives

$$\lambda_{\text{collar}} = \int dy w(y) [1 - \epsilon_{\mathbb{Z}_6}(y)] = 1 - \int dy w(y) \epsilon_{\mathbb{Z}_6}(y). \quad (33)$$

Scalar-conditioned reserve mean. The unconditioned reserve mean

$$\tau_q(Z_{6,r,C}) = P/24$$

does not automatically imply

$$\frac{\tau_q(\Pi_{r,C}^{\text{scal}} Z_{6,r,C})}{\tau_q(\Pi_{r,C}^{\text{scal}})} = P/24.$$

The finite-thickness branch uses the scalar-weighted reserve profile

$$\epsilon_{\mathbb{Z}_6}(y) = \frac{\tau(\Pi^{\text{scal}} Z_6 p_y)}{\tau(\Pi^{\text{scal}} p_y)}.$$

Thus the mean condition entering the survival average is

$$\int dy w(y) \epsilon_{\mathbb{Z}_6}(y) = P/24.$$

This is a scalar-weighted mean receipt. It follows on the uniform product-thickening branch, but not from co-registration alone.

If

$$\int dy w(y) \epsilon_{\mathbb{Z}_6}(y) = P/24, \quad (34)$$

then linearity of the integral gives the exact value

$$\lambda_{\text{collar}} = 1 - \frac{P}{24}. \quad (35)$$

The receipt alone fixes the coefficient; no constancy clause on $\epsilon_{\mathbb{Z}_6}(y)$ is consumed. Under the mean-count reading, with $\epsilon_{\mathbb{Z}_6}(y)$ the expectation of an $\mathbb{N}$ -valued occupancy and survival the zero-count event, Markov's inequality gives the assumption-minimal band

$$1 - \frac{P}{24} \leq \lambda_{\text{collar}} \leq 1. \quad (36)$$

Conditional bridge 6, exact finite-thickness coefficient. The imported exact-coefficient theorem is stated on the uniform product-thickening branch of the screen-microphysics paper:

$$\mathcal{A}_{r,C}^{\text{thick}} \cong \mathcal{E}_{r,C} \otimes \mathcal{T}_r(Y),$$

$$\tau_{q,r,C}^{\text{thick}} = \tau_{q,r,C}^{\mathcal{E}} \otimes \tau_r^Y,$$

$$Z_{6,r,C}^{\text{thick}} = Z_{6,r,C} \otimes 1_Y,$$

and, in condensed form,

$$\frac{\tau_{q,r,C}^{\mathcal{E}} \left(\Pi_{r,C}^{\text{scal}}(y) Z_{6,r,C} \right)}{\tau_{q,r,C}^{\mathcal{E}} \left(\Pi_{r,C}^{\text{scal}}(y) \right)} = \frac{P}{24}$$

for every scalar-active transverse slice y . Product trace alone is not enough. The scalar-active sampler must be reserve-unbiased.

Conditional argument. Under the presence reading of conditional bridge 5, on the uniform product-thickening branch, the imported theorem labeled `thm:exact-uniform-product-thickening-coefficient` gives

$$\lambda_{\text{collar}} = 1 - \frac{P}{24} = 0.9320429912748350 \dots \quad (37)$$

The scalar-weighted receipt fixes this value by linearity on its own; the uniform branch adds slice-wise constancy of the profile, which the exact value does not consume. Without the scalar-weighted receipt, the theorem-grade coefficient is

$$\lambda_{\text{collar}} = 1 - \int dy w(y) \epsilon_{\mathbb{Z}_6}(y).$$

Under the mean-count reading, with the scalar-weighted protected reserve mean $P/24$, Markov gives only

$$1 - \frac{P}{24} \leq \lambda_{\text{collar}} \leq 1.$$

Under the presence reading with the receipt,

$$\lambda_{\text{collar}} = 1 - \frac{P}{24} = 0.932042991274835 \dots \quad (38)$$

and by (23)

$$a_{0,\text{eff}} = 1.184737388 \times 10^{-10} \text{ m s}^{-2}. \quad (39)$$

Failure consequence. Without the scalar-weighted mean receipt, the coefficient is the integral (33). The exact value $1 - P/24$ is not available as a theorem-grade number.

10 Lensing And Relativistic Stress

The QUMOND equation specifies one weak-field potential and a virtual source. It does not specify the second metric potential, gravitational slip, a covariant stress tensor, or light propagation.

On a covariant-action route,

$$T_A^{ab} := - \frac{2}{\sqrt{-g}} \frac{\delta S_A}{\delta g_{ab}}$$

defines the metric stress and both scalar potentials. The relation $\Phi = \Psi$ holds only if the action and its solution imply it.

On a positive transported-stress route, finite collar packets define

$$J_A^a = \int v^a d\mathcal{E}_x(v), \quad T_A^{ab} = \int v^a v^b d\mathcal{E}_x(v),$$

with decomposition

$$T_A^{ab} = \epsilon_A u^a u^b + P_A h^{ab} + 2u^{(a} q_A^{b)} + \pi_A^{ab}. \quad (40)$$

Rotational isotropy can set $q_A^a = \pi_A^{ab} = 0$ locally, but a static extended configuration generally requires pressure or velocity dispersion. This positive stress cannot be identified with a generally signed QUMOND virtual density.

In Newtonian gauge,

$$ds^2 = a^2[-(1+2\Psi)d\eta^2 + (1-2\Phi)d\mathbf{x}^2], \quad k^2(\Phi - \Psi) = 12\pi G a^2 \sum_i (\rho_i + P_i)\sigma_i.$$

Consequently $\Phi_{\text{lens}} = (\Phi + \Psi)/2 = \Phi$ is not a prediction of this paper. Galaxy, cluster, and cosmological lensing require a complete stress, action, and boundary-condition calculation.

11 Minimal Covariant Continuation: The OPH-R1 Response Contract

One response functional governs every source. Let an observer clock scalar τ define

$$u_a = -\frac{\nabla_a \tau}{\sqrt{-\nabla_c \tau \nabla^c \tau}}, \quad h_{ab} = g_{ab} + u_a u_b, \quad D_a = h_a^b \nabla_b.$$

With $\rho_b = T_{ab}^{\text{SM}} u^a u^b / c^2$, introduce the baryonic source potential through the weak-field slice constraint

$$D_a D^a \Psi_b = 4\pi G \rho_b.$$

Its embedding as a propagated constraint of a stable hyperbolic covariant system, together with preferred-frame tests, is work in progress. Define

$$Y = \frac{D_a \Psi_b D^a \Psi_b}{a_{0,\text{OPH}}^2}, \quad \mathcal{S}_Q[\Psi_b] = D_a \left[(\nu_{\text{OPH}}(\sqrt{Y}) - 1) D^a \Psi_b \right].$$

The anomaly potential is the retarded solution of

$$\mathcal{L}_A \varphi_A = -c_A^2 \mathcal{S}_Q[\Psi_b], \quad \mathcal{L}_A = (u^a \nabla_a)^2 + 2\Gamma_{\text{rec}} u^a \nabla_a - c_A^2 D_a D^a + \mathcal{V}_A, \quad (41)$$

or

$$\varphi_A(x) = -c_A^2 \int_{J^-(x)} G_{\text{ret}}(x, x') \mathcal{S}_Q[\Psi_b](x') dV_{x'}.$$

Here $0 < c_A \leq c$, Γ_{rec} , and $\mathcal{V}_A$ have to come from the finite repair generator and operational clock, not independent fits to target systems.

Static normalization. For a time-independent solution with $\Gamma_{\text{rec}} = 0$ and $\mathcal{V}_A = 0$,

$$D^2 \varphi_A = \mathcal{S}_Q[\Psi_b], \quad \Phi = \Psi_b + \varphi_A,$$

which is exactly the QUMOND equation. The observer-foliated form of (19), with the clock included in the covariant data, provides the corresponding static action target.

Causality and conservation. The support of G_{ret} lies in $J^-(x)$. For $\Gamma_{\text{rec}} > 0$, damping exchanges stress with finite recipient degrees of freedom:

$$\nabla_a T_A^{ab} = Q^b, \quad \nabla_a T_R^{ab} = -Q^b, \quad \nabla_a (T_{\text{SM}}^{ab} + T_A^{ab} + T_R^{ab}) = 0.$$

A damped anomaly equation without recipient stress violates the Bianchi identity.

Unified source response. All baryonic sources enter Ψ_b . There is no internal/external switch or special Solar-System, binary, disk, or satellite exemption. Environmental response is determined by the retarded Green function, rate, and action.

Rate nonuniqueness. For the electric Weyl tensor $E_{ab} = C_{acbd}u^c u^d$, set

$$I_2 = E_{ab}E^{ab}, \quad I_3 = E_a{}^b E_b{}^c E_c{}^a, \quad \zeta = I_3/I_2^{3/2}.$$

The rotationally invariant homogeneous local family built only from E_{ab} is

$$\Gamma_{\text{rec}} = I_2^{1/4} F(\zeta).$$

The point-source-normalized choice $\Gamma_J = (I_2/6)^{1/4}$ is one constant-shape diagnostic, not a theorem or a cluster-data selector.

Acceptance gates. A physical OPH-R1 parent has to derive ν , a_0 , and the response kernel without target data; embed the slice constraint in a stable covariant hyperbolic action; recover conservative disk solutions; satisfy Cassini, preferred-frame, propagation, wide-binary, and satellite bounds; calculate lensing, cluster maps, FLRW evolution, and perturbations; close every exchange current; and pass a joint likelihood against Λ CDM, QUMOND, and halo alternatives. These gates are work in progress, so (41) is a completion contract rather than evidence for OPH.

12 Clusters And Transport

The following cluster equations diagnose candidate realizations of OPH-R1; they are not an independently selectable theory. A positive transported realization may reduce to a density and velocity field:

$$\partial_t \rho_{\mathcal{I}} + \nabla \cdot (\rho_{\mathcal{I}} \mathbf{v}_{\mathcal{I}}) = -\frac{\rho_{\mathcal{I}} - \rho_{\mathcal{I},\text{eq}}[\rho_b]}{\tau_{\text{rec}}}, \quad (42)$$

$$\partial_t \mathbf{v}_{\mathcal{I}} + (\mathbf{v}_{\mathcal{I}} \cdot \nabla) \mathbf{v}_{\mathcal{I}} = -\nabla \Phi, \quad (43)$$

$$\nabla^2 \Phi = 4\pi G(\rho_b + \rho_{\mathcal{I}}). \quad (44)$$

Velocity-spread branches use the moment equations of (40) instead of (42).

The repair-commit theorem gives the form of the relaxation term. A transition matrix has a dimensionless step eigenvalue. Physical relaxation uses its logarithmic decay together with the physical commit time. Let $r_\star$ be the largest nonstationary step eigenvalue in modulus, $g_{\text{step}} = 1 - r_\star$, and $\gamma_{\log} = -\log |r_\star|$. With physical commit time t_{commit} ,

$$\Gamma_{\text{rec}} = \frac{\gamma_{\log}}{t_{\text{commit}}}, \quad \tau_{\text{rec}} = \frac{t_{\text{commit}}}{\gamma_{\log}}. \quad (45)$$

The small-gap approximation is $\Gamma_{\text{rec}} = g_{\text{step}}/t_{\text{commit}} + O(g_{\text{step}}^2)$. The symbols r_* , g_{step} , γ_{log} , and Γ_{rec} denote different artifact types.

Bianchi consistency requires a repair-exchange stress:

$$\nabla_a(T_b^{ab} + T_{\mathcal{I}}^{ab} + T_R^{ab} + \dots) = 0. \quad (46)$$

In quasi-covariant notation,

$$\nabla_a T_{\mathcal{I}}^{ab} = -\Gamma_{\text{rec}}(\rho_{\mathcal{I}} - \rho_{\mathcal{I},\text{eq}})u_{\mathcal{I}}^b, \quad (47)$$

and T_R^{ab} carries the opposite exchange flux.

A useful timescale diagnostic is the constant-shape member of the OPH-R1 rate family. For the electric Weyl tidal tensor E_{ij} in a local free-fall frame, set

$$\Gamma_J = \left(\frac{E_{ij}E^{ij}}{6} \right)^{1/4}, \quad \tau_J = \Gamma_J^{-1}. \quad (48)$$

For a Newtonian point source,

$$E_{ij}E^{ij} = 6(GM/r^3)^2, \quad \tau_J = \sqrt{r^3/(GM)}.$$

The normalization is exact, but dimensionless tidal-shape invariants can change the rate away from point-source geometry. The table is a diagnostic, not a no-fit prediction. The numerical scale is:

Scale	$M/M_\odot$	r in kpc	τ_J in Gyr
Inner spiral disk	6.0×10^{10}	10	0.0608673
Outer spiral disk	6.0×10^{10}	30	0.316276
Dwarf scale	1.0×10^9	5	0.166692
Cluster core	1.0×10^{14}	300	0.244986
Massive cluster aperture	1.0×10^{15}	1000	0.471476

For $d_0 = 200$ kpc and $t = 0.2$ Gyr, the massive-cluster aperture row gives $d(t) = 130.859$ kpc.

The conditional physical-time conversion is

$$\Gamma_{\text{rec}}(x, t) = \Gamma_J(x, t), \quad t_{\text{commit}}(x, t) = \frac{\gamma_{\text{log}}(x, t)}{\Gamma_J(x, t)}. \quad (49)$$

The static RAR law does not determine this clock: any positive relaxation rate leaves the same equilibrium density. Derivation of the rate function from the finite repair generator, without cluster offsets as selectors, is work in progress.

For a one-mode merger offset,

$$d(t) = d_0 \exp(-t/\tau_{\text{rec}}), \quad \tau_{\text{rec}} = \frac{t}{\ln(d_0/d(t))}. \quad (50)$$

With $d_0 = 200$ kpc and $t = 0.2$ Gyr:

Retained offset	Required τ_{rec}
50 kpc	0.144 Gyr
100 kpc	0.288 Gyr
134 kpc	0.500 Gyr
150 kpc	0.696 Gyr
164 kpc	1.000 Gyr

A one-pole equilibrium diagnostic imposes

$$\tau_{\text{rec}} < \frac{T_{\text{set}}}{\ln(1/\epsilon)} \quad (51)$$

if the allowed tracking error after settling time T_{set} is ϵ . For $T_{\text{set}} = 5$ Gyr, the bounds are 2.17 Gyr, 1.67 Gyr, and 1.09 Gyr for 10%, 5%, and 1% tracking error.

13 CMB, BAO, Growth, And S_8

The linear cosmology contract has two distinct branches. A conserved homogeneous information-defect branch obeys

$$\rho'_A + 3\mathcal{H}\rho_A = 0, \quad \rho_A(a) = \rho_{A0}a^{-3}, \quad (52)$$

where primes denote conformal time and $\mathcal{H} = a'/a$. This branch behaves like a cold stress component if its kinetic measure is monokinetic and its homogeneous charge $Q_A = a^3\rho_A V_{\text{com}}$ is transported by an anomaly-current conservation receipt and selected by the source-only anomaly abundance receipt of Section 18.15. A source-localized CMI value without those receipts does not imply the a^{-3} scaling or the homogeneous amplitude.

No-go boundary for scalar parents. A scalar parent that emits only $\rho_A(a)$, $\rho_{A,\text{eq}}(a)$, and $B_A(k, a)$ is a source candidate, not a physical Boltzmann closure. It does not determine

$$w_A(a), \quad c_{s,A}^2(k, a), \quad \sigma_A(k, a), \quad Q_A^\mu, \quad \Gamma_{\text{rec}}(k, a),$$

and it does not by itself fix gauge-independent perturbations or the recipient stress needed when repair transfers energy-momentum. In the physical source contract B_A is built from the anomaly-frame baryon density

$$n_b^{(A)} = -u_{A\mu}J_b^\mu,$$

not from a gauge-fixed coordinate density. The physical object is a finite covariant collar-parent

$$\mathcal{P}_r[X, g] = (C_r, Z_r, A_r, R_r, G_r, \pi_r[X], L_r[X], Q_r, D_r),$$

where Z_r is a finite packet-state set split into anomaly and recipient channels, π_r is the convex equilibrium packet functional, L_r is the repair generator, Q_r records reaction channels, and D_r records any causal auxiliary response. The parent must emit the stress moments and source projections consumed by the Boltzmann solver; otherwise CMB/BAO/growth rows are diagnostics.

Finite covariant parent object. The theorem-grade parent is not a parent-shaped JSON object. At regulator r it is a quotient-visible finite kinetic package

$$\mathcal{P}_r[X, g] = (\mathcal{C}_r, g_r, e_r, U_r, Z_r, \omega_r, p_r, f_r, \Phi_r, \mathcal{R}_r, \mathcal{G}_r, \pi_r, \text{Read}_r),$$

with finite causal complex $\mathcal{C}_r$, local tetrads e_r , finite parallel transports U_r , packet sectors

$$Z_r = \bigsqcup_s Z_{s,r},$$

positive quadrature weights, occupations $f_{c,z} \geq 0$, and momenta obeying the declared $(- + ++)$ mass shell

$$g_{ab}p_z^a p_z^b = -m_z^2, \quad p_z^0 > 0.$$

The quotient $\mathcal{G}_r$ includes hidden carrier coordinates, equivalent packet labels, port relabelings, local-frame presentations, and accepted repair-schedule presentations. The readout

$$\text{Read}_r = (\text{StressRead}_r, \text{ExchangeRead}_r, \text{ScalarSourceMap}_r, \text{ClockRead}_r, \text{RepairTangent}_r)$$

must factor through that quotient.

Exact finite closure receipt. For each sector define

$$J_{s,c}^a = \frac{1}{V_c} \sum_{z \in Z_s} W_{c,z} p_z^a, \quad T_{s,c}^{ab} = \frac{1}{V_c} \sum_{z \in Z_s} W_{c,z} p_z^a p_z^b.$$

The simulator must reconstruct these moments from packet rows and emit residual ledgers

$$R_{s,c}^a = (D_r T_s)^a - Q_{s,c}^a, \quad R_{\text{total},c}^a = D_r T_{\text{total}}^a,$$

including internal-face cancellation, boundary flux, geometry/redshift work, external work, reaction four-momentum, and conserved charges. Producer booleans, zero-density labels, invented recipient rows, or scalar residuals are not stress closure. Promoting $\rho_A(a)$, $B_A(k, a)$, a certified $\Gamma_{\text{rec}}(k, a)$, or any associated primordial scalar input also requires a source-provenance DAG and a pooled-reducer receipt. A transition spectral number is $\gamma_{\text{repair step}}$ without active-fiber, physical-clock, and common-parent response receipts. Contradictory no-data-use metadata fails closed, and nonlinear quantities such as amplitudes, ranks, condition numbers, and isocurvature leakage are computed only after additive sufficient statistics are pooled globally.

For the source-only primordial bridge, this closed branch has the stricter contract

$$\nabla_a J_A^a = 0, \quad T_A^{ab} = \epsilon_A u^a u^b, \quad Q_A^a = 0. \quad (53)$$

It may enter the rank-one primordial theorem only when ρ_A is a function of the same source clock as the rest of the total stress. Transport of a supplied Q_A is not an OPH derivation of the homogeneous abundance.

A repair-exchange branch instead relaxes toward a finite-collar equilibrium source:

$$\rho'_A + 3\mathcal{H}\rho_A = -a\Gamma_{\text{rec}}(\rho_A - \rho_{A,\text{eq}}). \quad (54)$$

The factor $a\Gamma_{\text{rec}}$ is the conformal-time rate associated with the physical repair rate. Define

$$q_A = \frac{\rho_{A,\text{eq}}}{\rho_A}, \quad \delta_{A,\text{eq}}(k, a) = B_A(k, a) \delta_b(k, a). \quad (55)$$

For a pressureless scalar diagnostic with energy exchange parallel to the anomaly four-velocity, the Newtonian-gauge perturbations reduce to

$$\delta'_A = -\theta_A + 3\Phi' - a\Gamma_{\text{rec}} q_A (\delta_A - \delta_{A,\text{eq}}), \quad (56)$$

$$\theta'_A = -\mathcal{H}\theta_A + k^2\Psi. \quad (57)$$

These equations do not fix metric slip. Total slip depends on the OPH-R1 action and all baryonic, radiation, neutrino, anomaly, and recipient stresses.

For the same primordial bridge, the repair-exchange branch is admissible only with an explicit transfer receipt:

$$\nabla_a T_A^{ab} = Q_A^b, \quad \Xi_a^A = 0, \quad h^a_b Q_A^b = 0 \quad (58)$$

or with interval bounds on the two defects. The dark-sector continuation in this paper separates transport of a supplied homogeneous load from source selection of that load. Transport alone has label

PHYSICAL_PARENT_WITH_CONDITIONAL_ABUNDANCE.

A source-only homogeneous abundance requires

RHO_A_SOURCE_RECEIPT = RHO_A_TRANSPORT_RECEIPT $\wedge$ ANOMALY_ABUNDANCE_SOURCE_RECEIPT.

For primordial certification without that receipt, use

dark_continuation = OFF.

A supplied dark abundance may be used only as

dark_continuation = CONDITIONAL_SOURCE_STATE,

not as a derived OPH primordial prediction.

If the exchange branch is nonzero, the recipient sector is not optional. The finite parent must expose a recipient stress T_R^{ab} , a reaction channel, and equal-and-opposite exchange currents such that

$$\nabla_a(T_A^{ab} + T_R^{ab} + T_{\text{other}}^{ab}) = 0.$$

Without that explicit stress and current closure, a repair sink/source can mimic a scalar damping term but cannot be promoted to an Einstein-source prediction.

The finite scalar-load repair diagnostic gives

$$\gamma_{\text{repair step}} = 0.050717471.$$

It maps a cluster-scale relaxation window to the following commit times:

τ_{rec}	t_{commit}
0.3 Gyr	15.2152 Myr
0.5 Gyr	25.3587 Myr
1.0 Gyr	50.7175 Myr

For the same relaxation window, a simple background timing diagnostic gives:

τ_{rec}	$z = 0$	$z = 10$	$z = 1100$
0.3 Gyr	48.4092	2.36213	0.00205644
0.5 Gyr	29.0455	1.41728	0.00123387
1.0 Gyr	14.5228	0.708639	0.000616933

In this timing diagnostic, repair relaxation is small during acoustic physics and large in low-Hubble environments. A physical cosmology requires an OPH-R1 source artifact, Boltzmann implementation, and frozen full likelihoods for CMB TT/TE/EE, CMB lensing, BAO, weak lensing, redshift-space distortions, and S_8 .

14 Abundance And Kernel Targets

The static QUMOND law defines a diagnostic response around resolved baryonic inhomogeneity. A cosmological dark-matter branch requires OPH-R1 outputs:

$$\rho_A(a), \quad B_A(k, a), \quad \Gamma_{\text{rec}}(k, a), \quad w_A, \quad c_{s,A}^2, \quad \sigma_A, \quad Q_A^\mu. \quad (59)$$

The first is the homogeneous anomaly abundance. The second is the linear equilibrium-source kernel in

$$\delta_{A,\text{eq}}(k, a) = B_A(k, a)\delta_b(k, a). \quad (60)$$

The other entries close the stress tensor and exchange law.

14.1 Static FLRW Boundary

The nonlinear galaxy law cannot be inserted directly into FLRW perturbation theory. In an exactly homogeneous background there is no preferred baryonic acceleration vector. If one tries to Taylor-expand the static RAR law at $\mathbf{g}_b = 0$, the susceptibility is singular because $\nu_{\text{OPH}}(x) \sim (\lambda_{\text{collar}}\sqrt{x})^{-1}$. The static RAR branch therefore does not select a homogeneous anomaly density or a universal CMB kernel.

This is a useful boundary. It prevents a hidden fit from being smuggled into the galaxy law. The CMB branch has to declare its background anomaly charge and its perturbation kernel from OPH state selection or finite-collar microphysics.

14.2 Capacity-Saturated State Selection

One clean homogeneous benchmark is flat capacity saturation. More generally, the de Sitter screen capacity fixes H_{dS} ; the capacity itself is the SL-4 estimate read from measured Λ with the scale bridge (`./docs/STRANGE_LOOP_PRINCIPLES.md`), so H_{dS} is an input-carrying quantity rather than an OPH output. The object that would promote this capacity lane is generator G2 of the project consistency stack: constructing the readback map F and its contraction certificate. The Friedmann closure at $a = 1$ gives only the degenerate combination

$$\Omega_{A0} + \Omega_{K0} = 1 - \Omega_{\Lambda,\text{OPH}} - \Omega_{b0} - \Omega_{\nu0} - \Omega_{r0}, \quad \Omega_{\Lambda,\text{OPH}} = \left(\frac{H_{\text{dS}}}{H_0} \right)^2. \quad (61)$$

The displayed number becomes an Ω_{A0} value only after an independent curvature readout or an explicit flat-branch assumption $\Omega_{K0} = 0$. It does not prove flatness. The transported homogeneous dust branch is

$$\rho_A(a) = \rho_{A0}a^{-3}. \quad (62)$$

Equation (61) is a closure sum, not an anomaly-abundance prediction or a flatness proof. The static QUMOND law does not determine it. A source-only Ω_{A0} requires a finite-screen additive-load selector and an independent curvature label; both are work in progress.

Finite exports must therefore label

$$\Omega_{A,\text{source}} \in \{\text{INDEPENDENTOPH}, \text{EXTERNAL}, \text{RESIDUALDEFINITION}, \text{UNKNOWN}\}$$

and separately label the curvature source. A residual-defined Ω_A cannot pass a Friedmann-curvature independence audit.

The closed transported branch does give a theorem-grade current and charge. Let u_A^a be the branch four-velocity and let ρ_A be the coarse-grained information-defect load density. The load current is

$$J_A^a = \rho_A u_A^a. \quad (63)$$

On the closed branch,

$$\nabla_a J_A^a = 0, \quad Q_A[\Sigma] = \int_{\Sigma} J_A^a n_a d\Sigma \quad (64)$$

is independent of the homogeneous slice. In FLRW this gives

$$Q_A = a^3 \rho_A V_{\text{com}}, \quad \rho_A(a) = \frac{Q_A}{a^3 V_{\text{com}}}. \quad (65)$$

This proves transport of a supplied homogeneous load. It does not fix the amplitude Q_A . The source-only selector is the quotient expectation of the parent-emitted anomaly load on a release hypersurface,

$$\mathcal{E}_{A,r} = F_{A,r} \left(Q_{A,r}^{\text{rel}}, \mu_{A,r}^{\text{rel}}, P_{\star}, N_{\text{CRC}}, \mathfrak{R}_{A,r}^{\text{rel}}, \mathcal{P}_r \right), \quad \rho_A(a) = \frac{\mathcal{E}_{A,r}}{a^3 V_{\text{com},r}}. \quad (66)$$

Section 18.15 gives the finite selector and receipt. Without that source selector receipt, the abundance is cosmological state data or the flat capacity-saturated residual (61).

14.3 Environmental Kernel

Around a nonzero local background field $\bar{\mathbf{g}}_b$, write

$$x = \frac{|\bar{\mathbf{g}}_b|}{a_{0,\text{OPH}}}, \quad \mathbf{n} = \frac{\bar{\mathbf{g}}_b}{|\bar{\mathbf{g}}_b|}. \quad (67)$$

Linearizing the anomaly field $(\nu_{\text{OPH}} - 1)\mathbf{g}_b$ gives the susceptibility

$$A_{ij} = [\nu_{\text{OPH}}(x) - 1]\delta_{ij} + x\nu'_{\text{OPH}}(x)n_i n_j, \quad (68)$$

where

$$\nu'_{\text{OPH}}(x) = -\frac{\lambda_{\text{collar}} \exp[-\lambda_{\text{collar}} \sqrt{x}]}{2\sqrt{x}[1 - \exp(-\lambda_{\text{collar}} \sqrt{x})]^2}. \quad (69)$$

For a Fourier mode whose direction has cosine μ against $\mathbf{n}$, the density-response multiplier is

$$K_A^{(\rho)}(x, \mu) = \nu_{\text{OPH}}(x) - 1 + x\nu'_{\text{OPH}}(x)\mu^2. \quad (70)$$

An isotropic environmental average gives

$$\langle K_A^{(\rho)} \rangle = \nu_{\text{OPH}}(x) - 1 + \frac{x}{3} \nu'_{\text{OPH}}(x). \quad (71)$$

The contrast kernel used in perturbation equations is normalized by the equilibrium-background density ratio only after the response is diagonal in the calibrated physical mode basis:

$$B_A(k, a) = \frac{\bar{\rho}_b(a)}{\bar{\rho}_{A,\text{eq}}(a)} K_A^{(\rho)}(k, a). \quad (72)$$

Using $\bar{\rho}_A$ in the denominator would define the different variable $\delta\rho_{A,\text{eq}}/\bar{\rho}_A$ and would require red-eriving the external $q_A = \bar{\rho}_{A,\text{eq}}/\bar{\rho}_A$ factor in the perturbation equations. This paper uses the

equilibrium-contrast convention globally. At finite regulator the primary object is the response operator

$$(B_{A,r})_{jj'}(a) = \frac{\bar{\rho}_b(a)}{\bar{\rho}_{A,\text{eq}}(a)} \langle v_{rj}, D\rho_{A,\text{eq}}[\bar{\rho}_b] v_{rj'} \rangle, \quad (73)$$

where v_{rj} are the same calibrated comoving scalar modes used by the primordial source and transfer handoff. A scalar table $B_A(k_j, a)$ is promoted only if

$$\|[B_{A,r}, \Delta_{\text{com},r}]\| \leq \varepsilon_{\text{comm}}, \quad \sum_{j' \neq j} |(B_{A,r})_{jj'}|^2 \leq \varepsilon_{\text{off}}^2$$

on the resolved band. Otherwise the physical input is the mode-coupling matrix, not a diagonal function of k . In a realistic cosmological calculation, this local expression has to be integrated against the distribution of environmental fields,

$$K_A^{(\rho)}(k, a) = \int dx d\mu \Pi(x, \mu|k, a) [\nu_{\text{OPH}}(x) - 1 + x\nu'_{\text{OPH}}(x)\mu^2]. \quad (74)$$

This average is finite only under an explicit small-field support condition, for example

$$\int_0^\epsilon dx d\mu \Pi(x, \mu|k, a) x^{-1/2} < \infty \quad \text{for some } \epsilon > 0. \quad (75)$$

If Π has an exact atom at $x = 0$, the local nonzero-field kernel is not the correct FLRW response. The distribution Π is an OPH finite-collar output or an explicitly declared environmental closure. Exact FLRW has $x = 0$, so (70) does not provide a regular universal CMB density-response kernel by itself.

The implementation contract uses one parent finite-collar functional:

$$\rho_{A,\text{eq}}[X]c^2 = \frac{15}{8\pi^2\ell(X)^4} \int_{\mathcal{C}_X} d\mu_C R_C^{\text{src}}[\omega_C]. \quad (76)$$

For finite samples s with weights w_s , define

$$R(X) = \frac{\sum_s w_s R_s^{\text{src}}}{\sum_s w_s}, \quad (77)$$

$$K_A^{(\rho)}(k, a) = \frac{1}{\bar{\rho}_b} \frac{\partial \rho_{A,\text{eq}}}{\partial \delta_b(k, a)}, \quad (78)$$

$$B_A(k, a) = \frac{\bar{\rho}_b(a)}{\bar{\rho}_{A,\text{eq}}(a)} K_A^{(\rho)}(k, a). \quad (79)$$

The same evaluator emits the homogeneous equilibrium density, the linear density-response kernel, the settled galaxy source, and the cluster equilibrium source. A theorem-grade prediction requires the OPH collar measure $d\mu_C$, the finite-screen ensemble, and the small-field support condition. Fitting Π to CMB, weak-lensing, SPARC, or cluster data is excluded.

14.4 Physical Source And Kernel Promotion

The scalar tables used by Boltzmann codes are the last stage of a finite source construction. Let $Q_{r,a}$ be the finite quotient state space at regulator r and background scale factor a . A physical anomaly source is a quotient-visible C^1 functional

$$\mathcal{E}_{A,r} : Q_{r,a} \times \mathcal{A} \longrightarrow \mathbb{R} \quad (80)$$

such that

$$\bar{\rho}_{A,\text{eq},r}(a) = \mathcal{E}_{A,r}(q_{0,r}(a), a), \quad D\mathcal{E}_{A,r}|_{q_0} \text{ is the anomaly response used in } B_{A,r}. \quad (81)$$

For the fixed-reference modular-energy route one may write, on each frozen reference patch,

$$\mathcal{E}_{A,r}(q, a) = \rho_{A,\text{eq},r}^{(0)}(a) + \mathcal{N}_{r,a} \text{Tr}[(\rho_r(q) - \sigma_{r,a})K_{\sigma,r,a}^{\text{anom}}] + \mathcal{R}_{2,r}(q, a), \quad (82)$$

with $\mathcal{R}_{2,r}(q_0, a) = 0$ and $D\mathcal{R}_{2,r}|_{q_0} = 0$. Then

$$D\mathcal{E}_{A,r}|_{q_0}[v] = \mathcal{N}_{r,a} \text{Tr}[D\rho_r|_{q_0}[v] K_{\sigma,r,a}^{\text{anom}}].$$

The shift $K_{\sigma,r,a}^{\text{anom}} \mapsto K_{\sigma,r,a}^{\text{anom}} + cI$ leaves this tangent term invariant because $\text{Tr} D\rho_r[v] = 0$. The baseline $\rho_{A,\text{eq},r}^{(0)}$ is separate finite source data; a modular variation alone gives a response, not an absolute positive abundance. For a nonlinear-CMI route at an exact Markov reference, the first variation vanishes and the leading object is the Hessian. A nonzero linear response from CMI therefore requires a declared non-Markov background and its residual.

Let finite packet occupations satisfy

$$\mathcal{M}_{r,a} = \{f \in \mathbb{R}_{\geq 0}^{N_r} : A_{r,a}f = b_{r,a}\}, \quad Q_{r,a} = \mathcal{M}_{r,a}/\mathcal{G}_{r,a}.$$

At an interior background $f_0 > 0$,

$$T_{[f_0]}Q_{r,a} = \ker A_{r,a}/T_{f_0}(\mathcal{G}_{r,a} \cdot f_0). \quad (83)$$

Boundary backgrounds use the corresponding tangent cone. A symmetric $\pm\epsilon$ finite difference is admissible only when both perturbed states stay in the same admissible support. For a retraction $\mathcal{R}_{q_0}(0) = q_0$, $D\mathcal{R}_{q_0}(0) = I$, and a C^3 readout E_r ,

$$\frac{E_r(\mathcal{R}_{q_0}(\epsilon v)) - E_r(\mathcal{R}_{q_0}(-\epsilon v))}{2\epsilon} = DE_r[v] + O(\epsilon^2).$$

If the third directional derivative is bounded by M_v , the truncation error is at most $M_v\epsilon^2/6$.

Let $S_r : Q_{r,a} \rightarrow U_r$ be the delivered baryon-source map and $E_r : Q_{r,a} \rightarrow Y_r$ the anomaly source map. A physical response depending only on the delivered source exists exactly when

$$\ker DS_r|_{q_0} \subseteq \ker DE_r|_{q_0}. \quad (84)$$

Then there is a unique linear operator

$$B_{A,r} : \text{im } DS_r|_{q_0} \rightarrow Y_r$$

such that $DE_r|_{q_0} = B_{A,r}DS_r|_{q_0}$. This is the lift-independence condition: perturbing a collar width, packet occupation, or port current gives the same kernel only if those microscopic lifts deliver the same physical source tangent.

The delivered source should initially be the vector

$$\mathcal{S}_b = (\Delta_b^{(A)}, \Theta_b^{(A)}, \delta T_b^{[\tau]}, \delta x_e^{[\tau]}, \Phi_{\text{GI}}, \Psi_{\text{GI}}, \dots),$$

and the anomaly output the vector

$$\mathcal{Y}_A = (\delta\rho_A, \delta p_A, q_A, \pi_A, \delta Q_A, F_A, \dots).$$

A scalar $B_A(k, a)$ is licensed only when non-density columns are below the declared frozen residual threshold. If geometry feedback is present, the Boltzmann handoff normally receives the bare partial derivatives (E_b, E_g) , while the dressed static gain is

$$B_{\text{dressed}} = (I - E_g G_A)^{-1} (E_b + E_g G_b).$$

Passing the dressed gain into a solver that separately evolves metric feedback would count gravity twice.

The physical repair rate has the same source discipline. An equilibrium sampler kernel does not determine physical time. The finite parent must provide native rates $q_u(x, y)$ and a physical clock,

$$(L_u f)(x) = \sum_{y \neq x} q_u(x, y) [f(y) - f(x)].$$

The generator descends to the quotient iff, for any two representatives $\pi(x) = \pi(x')$,

$$\sum_{y: \pi(y)=q'} q_u(x, y) = \sum_{y: \pi(y)=q'} q_u(x', y) \quad \text{for every quotient state } q'.$$

Only this native generator, together with quotient lumpability and physical rate units, can promote Γ_{rec} . Static settled-state response, step response, and the integrated retarded kernel must agree in the common parent model before a one-pole reduction is emitted.

The physical kernel receipts are therefore

```
COMMON_SOURCE_FUNCTIONAL_RECEIPT
ADMISSIBLE_SOURCE_TANGENT_RECEIPT
CONSTRAINT_PRESERVING_RETRACTION_RECEIPT
SOURCE_VECTOR_SUFFICIENCY_RECEIPT
B_A_SOURCE_LIFT_INDEPENDENCE_RECEIPT
SOURCE_DESIGN_IDENTIFIABILITY_RECEIPT
FINITE_DIFFERENCE_ORDER_RECEIPT
C1_REFINEMENT_RECEIPT
ORDER_OF_LIMITS_RECEIPT
NATIVE_REPAIR_GENERATOR_RECEIPT
PHYSICAL_RATE_UNIT_RECEIPT
QUOTIENT_LUMPABILITY_RECEIPT
STATIC_DYNAMIC_RESPONSE_CONSISTENCY_RECEIPT
```

The simulator may emit diagnostic candidates before these receipts pass, but those candidates are not physical B_A , $\rho_{A,\text{eq}}$, or Γ_{rec} kernels.

15 Likelihood Contracts

15.1 SPARC Disk-Potential Contract

The publication-grade SPARC test must solve the conservative disk equation

$$\nabla^2 \Phi = \nabla \cdot \left[\nu_\lambda \left(\frac{|\nabla \Phi_b|}{a_{0,\text{OPH}}} \right) \nabla \Phi_b \right] \quad (85)$$

on an axisymmetric baryonic disk model. The hierarchy must marginalize or profile distance, inclination, disk and bulge mass-to-light ratios, gas scale, intrinsic scatter, velocity covariance,

quality cuts, and grid convergence. The unit branch and the $\mathbb{Z}_6$ presence branch are fixed before this test; SPARC is used as a comparison, not as the selector for λ_{collar} . The equation is evaluated on selected SPARC galaxies as a diagnostic, not as an applicability rule. Its universal extension is excluded by Cassini, and no physical scope follows without a single source-derived OPH-R1 kernel.

15.2 Cluster Map Contract

The cluster contract requires gas hydrodynamics, stellar and galaxy tracers, OPH-R1 stress, lensing projection, X-ray emissivity, SZ pressure, merger-age priors, and covariance-aware map likelihoods. It supplies no prediction; the parent stress, clock, metric potentials, and exchange terms are unspecified.

15.3 Boltzmann Contract

The Boltzmann module exposes

$$\bar{\rho}_A(a), \quad \bar{\rho}_{A,\text{eq}}(a), \quad w_A(a), \quad c_{s,A}^2(k,a), \quad \sigma_A(k,a), \quad Q_A^\mu, \quad B_A(k,a), \quad \Gamma_{\text{rec}}(k,a),$$

with an explicitly declared external neutrino-mass scenario in the same run. The rejected weighted-cycle tuple may appear only as a labeled compare-only scenario; it may not be a default, a source input, or a promotion path. Every dark-sector entry must be derived from a frozen finite covariant parent artifact with no CMB, BAO, lensing, SPARC, or cluster fit in that source step. It must recover the Λ CDM cold-component limit when the exchange and stress corrections are turned off, reproduce the compressed diagnostic rows only as a plumbing check, and then run the full CMB TT/TE/EE, CMB lensing, BAO, supernova, weak-lensing, RSD, and S_8 likelihoods under the same nuisance model used for Λ CDM and $w_0 w_a$. The physical likelihood run is frozen by immutable hashes for the source artifact, Boltzmann solver, and likelihood configuration before the data comparison. The same run must expose official likelihood and CDM-limit reductions at global scope. Shard-local `any()` rollups do not promote the source artifact.

16 Numerical Galaxy Comparison

The OPH capacity benchmark used here is

$$a_{0,\text{OPH}} = 1.029186271 \times 10^{-10} \text{ m s}^{-2}. \quad (86)$$

The value of $a_{0,\text{OPH}}$ is built from the measured H_0 and Λ through the capacity de Sitter branch and the SI clock bridge. Under the strange-loop principle set these are SL-4/SL-5 inputs (`./docs/STRANGE_LOOP_PRINC`) so $a_{0,\text{OPH}}$ inherits two measured cosmological quantities. Every comparison of $a_{0,\text{OPH}}$ or $a_{0,\text{eff}}$ against the empirical acceleration scale in this section is therefore a consistency display between measured inputs, not an OPH prediction. The common empirical reference is

$$a_0^{\text{emp}} = 1.2 \times 10^{-10} \text{ m s}^{-2}. \quad (87)$$

The $\mathbb{Z}_6$ presence branch gives

$$\lambda_{\text{collar}} = 0.932042991, \quad a_{0,\text{eff}} = 1.184737388 \times 10^{-10} \text{ m s}^{-2}. \quad (88)$$

Quantity	Value	Comparison
$a_{0,\text{OPH}}$	$1.029186271 \times 10^{-10}$	−8.070% versus best same-function RAR
$a_{0,\text{OPH}}$	$1.029186271 \times 10^{-10}$	−14.234% versus empirical reference
$a_{0,\text{eff}}, \mathbb{Z}_6$ presence	$1.184737388 \times 10^{-10}$	+5.825% versus best same-function RAR
$a_{0,\text{eff}}, \mathbb{Z}_6$ presence	$1.184737388 \times 10^{-10}$	−1.272% versus empirical reference

These rows are consistency displays: $a_{0,\text{OPH}}$ carries measured H_0 and Λ , and an exact hit on the common empirical a_0 would require about 9% more protected inactive reserve than the $P/24$ behind $\lambda_{\text{collar}} = 1 - P/24$ (see Static Parameter Status).

The detailed tables below evaluate the mean-count chart $\lambda_{\text{collar}} = e^{-P/24}$, with $a_{0,\text{eff}} = 1.179018696 \times 10^{-10} \text{ m s}^{-2}$. They do not evaluate the presence chart $\lambda_{\text{collar}} = 1 - P/24$ and cannot distinguish the two factors inside $a_{0,\text{eff}}$.

16.1 SPARC RAR

At $\lambda_{\text{collar}} = 1$ the response function $\nu_{\text{OPH}}(x)$ of (17) coincides with the McGaugh–Lelli–Schombert empirical interpolating function, which was fitted to the SPARC radial acceleration relation [8]. The SPARC comparisons below therefore score a SPARC-fitted functional form against SPARC itself: they are not held-out tests, and per the sensitivity audit they mostly test normalization rather than shape. Using the public SPARC RAR data [7, 8], the local diagnostic gives:

Scenario	a_0 in m s^{-2}	RMS dex	Binned RMS dex
OPH unit	$1.029186271 \times 10^{-10}$	0.134208	0.026169
$\mathbb{Z}_6/\text{Poisson}$	$1.179018696 \times 10^{-10}$	0.132834	0.015760
Empirical reference	$1.200000000 \times 10^{-10}$	0.132913	0.016164
Best same-function	$1.119530397 \times 10^{-10}$	0.132947	0.017428

All rows share the SPARC-fitted functional form and are non-held-out; the comparison mostly tests normalization. An exact hit on the common empirical a_0 would require about 9% more protected reserve than $P/24$.

16.2 Rotation Diagnostics

The fixed stellar mass-to-light diagnostic and the first nuisance-profiled diagnostic give:

Scenario	Fixed-M/L RMS	Profiled-M/L χ^2/pt	Profiled RMS
OPH unit	23.366208	1.341948	12.420181
$\mathbb{Z}_6/\text{Poisson}$	22.868866	1.390941	12.590203
Empirical reference	22.842145	1.400930	12.624079

Units for the RMS columns are km s^{-1} . These rows inherit the SPARC-fitted functional form and the measured-input $a_{0,\text{OPH}}$; they are non-held-out consistency displays that mostly test normalization.

16.3 Systematic Likelihood Scaffold

A stronger local scaffold profiles stellar M/L, distance, inclination, and gas scale per galaxy using the SPARC sample table [6, 7]. It uses $Q \leq 2$, inclination at least 30° , and 3168 rotation-curve rows.

Scenario	χ^2/pt	χ^2/dof	RMS	Median disk M/L
OPH unit	1.016589	1.275672	11.249541	0.519102
$\mathbb{Z}_6$ /Poisson	1.021093	1.281324	11.276832	0.497435
Empirical reference	1.023261	1.284045	11.285518	0.493865

The table compares combined $a_{0,\text{eff}}$ charts and cannot validate the $\mathbb{Z}_6$ reserve separately. Its SPARC functional form is non-held-out, and the rows are dominated by normalization and nuisance choices.

16.4 Fixed-Branch RAR And BTFR Results

The aggregate RAR calculation is a fixed-branch test. Without refitting either OPH branch to the 2,693 public RAR rows from 153 galaxies, the conditional $\mathbb{Z}_6$ scale gives 0.1328335 dex RMS, while the unit endpoint gives 0.1342082 dex. The effective scale that minimizes the same table’s diagonal chi-square proxy has RMS 0.1329467 dex. These are non-held-out diagnostics rather than a blind prediction: $a_{0,\text{OPH}}$ is a capacity/cosmology benchmark, exact $\lambda_{\text{collar}} = 1 - P/24$ is conditional, and the aggregate rows share distance, inclination, and stellar-population systematics. A full nuisance-aware SPARC analysis independently reports

$$a_0 = (1.19 \pm 0.04_{\text{stat}} \pm 0.09_{\text{sys}}) \times 10^{-10} \text{ m s}^{-2},$$

placing the conditional $\mathbb{Z}_6$ value near its central result [10].

An error-aware BTFR fit uses orthogonal maximum likelihood with both-axis errors and intrinsic perpendicular scatter, giving

$$s_{\text{BTFR}} = 3.845654 \pm 0.085817,$$

reproducing the published 3.85 ± 0.09 result [9]. The OPH deep-limit slope 4 is 1.80σ high and lies within the published systematic range 3.5–4.0. At the declared 100 km s^{-1} pivot, however, the fixed-slope fit gives $\log_{10}(M_b/M_\odot) = 9.670792 \pm 0.020827$, whereas the conditional $\mathbb{Z}_6$ presence normalization predicts 9.803454. The data-minus-model offset is therefore -0.132662 dex, or -6.37σ using only the fit’s statistical normalization error. That pull omits a global stellar mass-to-light shift, distances, inclinations, sample selection, or whether every reported V_f has reached the asymptotic regime. The 123-galaxy BTFR table is drawn from SPARC, so it is a separate observable on an overlapping sample rather than an independent galaxy population.

These aggregate checks evaluate the algebraic RAR proxy. They do not replace the conservative disk solve of the likelihood contract. In non-spherical geometry $\nabla\nu \times \mathbf{g}_b$ need not vanish, so a publication-grade OPH test must expose the curl-field residual pattern rather than assume the pointwise algebraic relation.

16.5 Matched RAR–BTFR Closure Test

For each galaxy with an outer low-acceleration RAR point and an asymptotic or corrected V_f , define

$$r_{R,i} = \log g_{\text{obs},i} - \frac{1}{2}(\log g_{b,i} + \log a_{0,\text{eff}}),$$

$$r_{B,i} = \log M_{b,i} - 4 \log V_{f,i} + \log(Ga_{0,\text{eff}}), \quad \boxed{C_i = r_{B,i} + 2r_{R,i} - \epsilon_{\text{geom},i}}.$$

At leading order, C_i cancels common distance, inclination or velocity, baryonic-mass, and $a_{0,\text{eff}}$ shifts. The correction $\epsilon_{\text{geom},i}$ contains the full disk potential, finite-radius BTFR departure, pressure support, and covariance. The joint equilibrium law predicts a zero-centered C_i distribution under a held-out or leave-one-galaxy-out test with common nuisance priors.

16.6 Cassini Applicability Stress

If the conservative settled-galaxy equation is extrapolated universally to a Solar source embedded in the measured Galactic external field, it predicts an external-field quadrupole. The angular integral reproduces the published spherical benchmark and gives

$$Q_2^{\mathbb{Z}_6} \simeq 3.62 \times 10^{-26} \text{ s}^{-2}, \quad Q_2^{\lambda=1} = 3.40219 \times 10^{-26} \text{ s}^{-2}.$$

Against the Cassini summary $Q_2 = (1.6 \pm 1.8) \times 10^{-27} \text{ s}^{-2}$, these are fixed-input residuals of approximately 19.2σ and 18.01σ , respectively [11]. Adding only the linearized Gaia uncertainty on the Galactic acceleration reduces the $\mathbb{Z}_6$ pull to about 10.71σ , a strongly negative result [12].

This falsifies the natural universal/full-source extension of the displayed static equation, not the recovered OPH core. A system-specific switch is a fitted continuation. OPH-R1 instead requires one source-derived response kernel for Solar-System and galaxy observables.

16.7 Environmental Response Constraint

For a source-free interior, Gauss’s law gives

$$\oint_S \mathbf{g}_{\text{ext}} \cdot d\mathbf{S} = 0.$$

This constrains the trace of the tidal Hessian; it does not imply

$$D_i D_j \Phi_{\text{ext}} = 0.$$

A harmonic external potential can carry a quadrupole and higher multipoles, so zero external flux does not remove the Cassini tide. Enclosed-mass activation also fails for disks: a baryonic ring produces a nonzero field at interior points despite zero enclosed mass for a centered cut. The closed-cut predicate therefore supplies no Solar-System, disk, wide-binary, or satellite prediction.

All environmental observables belong to OPH-R1. Generalized QUMOND permits size- or timescale-dependent screening, but screening alone is not OPH-specific [21]. The source derivation and one frozen cross-system kernel are the discriminating content.

The empirical requirements are a direct Cassini Q_2 likelihood; the matched RAR–BTFR statistic; one frozen wide-binary and satellite population likelihood [24, 25]; a registered redshift-dependent RAR prediction tested against intermediate-redshift data [23]; and lensing and cluster maps derived from the same action and response kernel. These predictions are work in progress.

16.8 Implementation Work Packages

The empirical program follows the logical dependency order:

1. verify an axisymmetric and three-dimensional QUMOND solver against independent implementations, including curl and grid residuals;
2. evaluate the matched C_i statistic in a hierarchical SPARC likelihood with common nuisance priors;
3. implement (41) with causal support, stability bounds, and recipient exchange closure;
4. solve the same parent in the Sun–Galaxy environment and evaluate the Cassini Q_2 likelihood;

5. run one frozen wide-binary and satellite population model including triples, projection, disruption, and selection;
6. derive both metric potentials and generate galaxy and cluster lensing maps without imposing $\Phi = \Psi$;
7. emit background abundance, stress, exchange, perturbation kernels, and a CDM-limit regression for CAMB or CLASS;
8. register $a_{0,\text{app}}(z)$, scatter, and selection dependence;
9. run a reproducible joint likelihood with fixed data and source hashes against Λ CDM, QUMOND, and halo alternatives.

16.9 Static Parameter Status

The static interpolation is empirical QUMOND, and

$$(a_0, \lambda) \mapsto (a_0/\lambda^2, 1)$$

leaves every static prediction invariant. Coefficient decompositions are coordinate charts of one observable scale $a_{0,\text{eff}}$, not independent static tests. The values of $P/24$, $1 - P/24$, and the displayed accelerations are arithmetic diagnostics; they do not close the source derivation.

Universal QUMOND fails Cassini, the QUMOND source can be signed, the invariant tidal-rate family contains an arbitrary shape function, and relativistic slip requires a complete action. OPH-R1 source emission, lensing, clusters, and cosmological likelihoods are work in progress.

17 Claim Boundaries

Pressure point	Statement	Status
Finite source	A collar remainder becomes a source only after matching, tomography, quotient, and refinement receipts.	Conditional program.
Static galaxy law	The equation is QUMOND with an empirical RAR interpolation; the Poisson theorem holds on its model class and flux recovery is a premise.	Non-held-out continuation.
Static parameters	Only $a_{0,\text{eff}} = a_{0,\text{OPH}}/\lambda_{\text{collar}}^2$ is observable.	Exact non-identifiability.
Solar System	Universal QUMOND violates the Cassini quadrupole constraint.	Excluded.
Closed-cut screening	Zero flux does not remove tides, and enclosed-mass activation fails for disks.	Invalid.
Lensing	QUMOND and a scalar source do not fix relativistic slip.	No physical prediction.
Clusters	Offset equations and clocks are diagnostics without a source-derived parent and map likelihood.	No physical prediction.
Cosmology	Abundance, stress, response kernels, and joint likelihoods are work in progress.	No physical prediction.

Pressure point	Statement	Status
OPH-R1	One retarded covariant response contract includes a QUMOND static limit and recipient stress.	Defined completion target.

The supported claim is:

OPH supplies a finite self-reading source architecture and a falsifiable program for deriving gravitational response from patch repair. Its static implementation is QUMOND with one identifiable acceleration scale. Universal QUMOND is excluded by Cassini. A physical dark sector requires one source-derived OPH-R1 parent for galaxies, Solar-System observables, binaries, lensing, clusters, and cosmology; that derivation is work in progress.

Unsupported claim:

OPH has derived dark matter, uniquely explained the RAR, or solved MOND's cluster and cosmological problems.

18 Proof Ledger

Target	OPH closure	Numerical state
Galaxy source	Minimal scalar recoverability defect $R_C = I(A : D B)$.	Conditional source target.
Quotient-edge locality	Scalar repair opportunities are edge-center cut-register events on the physical quotient.	Conditional bridge.
Coherent material source lift	A nondegenerate OPH-coherent material source $S_{\text{coh}}^{\text{can}}$ descends to the quotient and is forced into the same edge-center scalar channel by Scalar Edge-Center Exhaustion.	Theorem-grade scalar-channel identity in the screen microphysics and χ_ν papers; finite stress-parent, abundance, and Boltzmann gates are separate obligations.
Finite channel bridge	One finite channel carries both the scalar-slot record panel and the collar-slice reserve panel, so the source generator and collar survival factor are composed on the same indexed family.	Bookkeeping bridge; nature-instantiation and G9 numerical calibration are work in progress.
Finite thickness	The presence reading plus the scalar-weighted reserve mean licenses the exact coefficient by linearity; otherwise the profile integral and, under the mean-count reading, the Markov band apply.	Conditional imported theorem.
Relativistic stress	QUMOND virtual density and positive packet stress are distinct; slip requires a covariant action or metric-response theorem.	No lensing theorem.
Repair relaxation	Fixed repair-commit chain gives $\tau_{\text{rec}} = t_{\text{commit}}/\gamma \log$.	Conditional packet theorem and clock candidate.
CMB and growth	Nonzero-environment kernel is explicit; exact FLRW static RAR kernel is blocked by a no-go theorem.	Finite-parent Boltzmann contract.
Abundance history	Flat capacity state selection gives a CDM-like residual; the closed transported branch has a conserved load current and charge.	Amplitude selector not emitted by this paper.
Linear transfer	The perturbation equation has a relaxation transfer solution between CDM-like and tracking limits.	Requires evaluated $B_A(k, a)$, recipient stress and exchange-current closure for nonzero exchange, physical-clock receipts for any promoted Γ_{rec} , and frozen hashes.
Parent functional	The finite collar expectation of the positive repair defect emits $\rho_{A,\text{eq}}$ and B_A if its FLRW limit is regular.	Scalar source target, not full stress closure.
Repair matrix	A reversible Metropolis kernel on packet states gives a concrete K .	Conditional on declared packet schema.
Repair rate family	$\Gamma = I^{1/4} E(I/(I^{3/2}))$; the constant shape Jacobian	Source derived, E is

18.1 Positive Galaxy Source

Theorem target 0A. The scalar carried charge is fixed by the exact modular-defect identity, not by a separate uniqueness principle. For a faithful finite collar state ω_{ABD} ,

$$-\langle K_{ABD} - K_{AB} - K_{BD} + K_B \rangle_\omega = I_\omega(A : D|B).$$

After reference subtraction on an exact Markov collar, this defines the canonical carried modular charge $M_C = I_\omega(A : D|B)$ in nats. A quotient-invariant collar allocation may sum these charges over a proper small-ball cover only when it passes the source-localization saturation receipt connecting the same carried operator to the finite-cutoff Einstein remainder.

Explicit premise. The galaxy-scale finite-stage modular nonadditivity is the carried modular charge appearing in the fixed-cap small-ball remainder, with hidden carrier duplicates, port belongings, repair schedules, and refinement-equivalent covers quotiented out.

Conditional argument. Conditional mutual information is nonnegative, so the source-localized carried charge is nonnegative. Equation (6) then gives a nonnegative rest-mass density after the source-localization receipt passes. The scalar charge alone does not determine T_A^{ab} ; packet factorization, causal directions, allocation, rest frame, and transport law are required to lift it to a stress tensor. $\square$

Proof obligation. Prove the operator-level source-localization relation between the finite-stage small-ball modular remainder and the quotient-invariant collar charge, then prove the positive packet allocation and covariant causal packet lift.

Failure consequence. Without source localization, $I(A : D|B)$ is only a recoverability and collar-diagnostic number. Without the packet lift, it cannot support lensing, CMB, growth, or conservation claims as a physical stress source.

18.2 Quotient-Edge Scalar Opportunity

Imported theorem. The fixed-cutoff uniqueness statement is discharged in the microphysics paper by

$$\text{Scal}_{r,C} = \mathcal{E}_{r,C}$$

on the edge-center scalar-completeness branch, and by the following imports:

$$\Pi_{r,C}^{\text{scal}} \in \mathcal{E}_{r,C}, \quad Z_{6,r,C} \in \mathcal{E}_{r,C}, \quad [\Pi_{r,C}^{\text{scal}}, Z_{6,r,C}] = 0.$$

The relevant theorem labels in the microphysics paper are

`thm:scalar-activation-edge-center`, `thm:scalar-z6-same-register-commute`, `thm:scalar-channel-ex`

For coherent material support the imported source is

$$S_{\text{coh}}^{\text{can}} = \mathbf{1}_{\text{self-read}} R_U P_U C_U \in \mathcal{E}_{r,C}$$

on the nondegenerate branch. This import fixes the scalar channel used by the χ_ν coefficient and does not supply a finite covariant dark-stress parent.

Branch boundary. The import is conditional on the microphysics assumptions labeled `ass:edge-center-scalar` and `ass:scalar-activation-channel-completeness`. If either fails, then scalar activation may require an additional quotient-visible scalar carrier, and the $\mathbb{Z}_6$ reserve cannot determine λ_{collar} without an independent co-registration theorem.

18.3 Uniform Finite-Thickness Reserve

Correct exact-value premise. The imported exact-coefficient theorem is stated on the uniform product-thickening branch of the microphysics paper:

$$\mathcal{A}_{r,C}^{\text{thick}} \cong \mathcal{E}_{r,C} \otimes \mathcal{T}_r(Y),$$

$$\tau_{q,r,C}^{\text{thick}} = \tau_{q,r,C}^{\mathcal{E}} \otimes \tau_r^Y,$$

$$Z_{6,r,C}^{\text{thick}} = Z_{6,r,C} \otimes 1_Y,$$

and, in condensed form,

$$\frac{\tau_{q,r,C}^{\mathcal{E}} \left(\Pi_{r,C}^{\text{scal}}(y) Z_{6,r,C} \right)}{\tau_{q,r,C}^{\mathcal{E}} \left(\Pi_{r,C}^{\text{scal}}(y) \right)} = \frac{P}{24}$$

for every scalar-active transverse slice y . Product trace alone is not enough. The scalar-active sampler must be reserve-unbiased.

Imported theorem. Under the presence reading of the reserve profile, on the uniform product-thickening branch, the microphysics theorem labeled `thm:exact-uniform-product-thickening-coefficient` gives

$$\lambda_{\text{collar}} = 1 - \frac{P}{24} = 0.9320429912748350 \dots$$

The scalar-weighted receipt fixes this value by linearity on its own; the uniform branch adds slice-wise constancy of the profile, which the exact value does not consume. Without the scalar-weighted receipt, the theorem-grade coefficient is

$$\lambda_{\text{collar}} = 1 - \int dy w(y) \epsilon_{\mathbb{Z}_6}(y).$$

Under the mean-count reading, with the scalar-weighted protected reserve mean $P/24$, Markov gives only

$$1 - \frac{P}{24} \leq \lambda_{\text{collar}} \leq 1.$$

18.4 Relativistic Stress Compatibility

A positive finite-collar kinetic measure, when it exists, defines

$$T_A^{ab} = \int v^a v^b d\mathcal{E}_x(v).$$

This tensor is future-positive, whereas the QUMOND source (25) is generally signed. They cannot be identified pointwise.

A covariant-action route derives metric stress and slip by varying the OPH-R1 action. A positive-stress route instead needs pressure, internal stress, exchange, or nonlocal coupling to reproduce the static metric response. No-slip follows only from a complete parent with negligible

total anisotropic stress and suitable boundary conditions. Monokinetic dust is not a generic static halo because separately conserved dust cannot remain static in an inhomogeneous potential.

The required parent action or packet stress, stationary solution, all-sector anisotropic-stress ledger, and lensing map are work in progress. A scalar density row or QUMOND potential supplies no $\Phi = \Psi$ theorem.

18.5 Repair Relaxation Scale

Theorem 0E. Let K be the fixed-cutoff repair-commit transition matrix for one local coarse cell at fixed baryonic source. If K is finite, irreducible, and aperiodic, with largest nonstationary eigenvalue r_* in modulus, then

$$\tau_{\text{rec}} = \frac{t_{\text{commit}}}{-\log |r_*|}. \quad (89)$$

If every allowed repair move has probability at least $q_{\min} > 0$ and the comparison chain has conductance Φ_K , then

$$1 - |r_*| \geq \frac{1}{2} q_{\min} \Phi_K^2. \quad (90)$$

Proof. A finite irreducible aperiodic chain has a unique stationary state. Every nonstationary mode decays as $|r_j|^n$ in repair-step time. One physical commit step takes t_{commit} , so physical relaxation has rate $-\log |r_*|/t_{\text{commit}}$. The conductance bound follows from the standard Cheeger comparison for the reversible comparison chain and the lower move probability $q_{\min}$. $\square$

Theorem 0E'. Equilibrium data do not select physical repair kinetics. Let P be an irreducible finite transition matrix with stationary law $\pi P = \pi$. For any $0 < \alpha \leq 1$, define

$$P_\alpha = (1 - \alpha)I + \alpha P.$$

Then $\pi P_\alpha = \pi$. If r_j is an eigenvalue of P , the corresponding eigenvalue of P_α is

$$r_j(\alpha) = 1 - \alpha + \alpha r_j,$$

and the physical decay rate at commit time $\Delta t_{\text{physical}}$ is

$$\Gamma_j(\alpha) = -\frac{\log |1 - \alpha + \alpha r_j|}{\Delta t_{\text{physical}}}.$$

Thus the same equilibrium law, $\rho_{A,\text{eq}}$, B_A , and detailed balance can support different kinetics. A sampler built from an equilibrium functional is a valid equilibrium evaluator only after its proposal law, attempt frequency, causal neighborhood, active fiber, and physical clock are declared as part of the OPH repair generator. $\square$

18.6 QUMOND Environmental Diagnostic

Linearizing the universal QUMOND equation around a nonzero field gives

$$K_Q^{(\rho)}(x, \mu) = \nu(x) - 1 + x\nu'(x)\mu^2,$$

where $x = |\bar{\mathbf{g}}_b|/a_0$ and μ is the cosine between the Fourier mode and background field. This follows from

$$\delta g_{A,i} = ([\nu(x) - 1]\delta_{ij} + x\nu'(x)n_i n_j) \delta g_{b,j},$$

with

$$\nu'(x) = -\frac{\lambda_{\text{collar}} \exp[-\lambda_{\text{collar}} \sqrt{x}]}{2\sqrt{x}[1 - \exp(-\lambda_{\text{collar}} \sqrt{x})]^2}.$$

This is a derivative of the Cassini-excluded universal static branch, not a physical OPH environmental kernel. OPH-R1 has to supply a retarded, gauge-consistent kernel and reproduce this expression only where its quasi-static limit applies.

18.7 Background Abundance Green Function

Theorem 0G. Let the background anomaly density obey

$$\rho'_A + 3\mathcal{H}\rho_A = -a\Gamma_{\text{rec}}(\rho_A - \rho_{A,\text{eq}}). \quad (91)$$

Define the comoving densities

$$R_A = a^3 \rho_A, \quad R_{A,\text{eq}} = a^3 \rho_{A,\text{eq}}. \quad (92)$$

Then

$$R_A(\eta) = W(\eta, \eta_i) R_A(\eta_i) + \int_{\eta_i}^{\eta} d\eta' W(\eta, \eta') a(\eta') \Gamma_{\text{rec}}(\eta') R_{A,\text{eq}}(\eta'), \quad (93)$$

where

$$W(\eta, \eta') = \exp \left[- \int_{\eta'}^{\eta} d\tilde{\eta} a(\tilde{\eta}) \Gamma_{\text{rec}}(\tilde{\eta}) \right]. \quad (94)$$

Proof. Multiplying the background equation by a^3 gives

$$R'_A = -a\Gamma_{\text{rec}}(R_A - R_{A,\text{eq}}). \quad (95)$$

This is a first-order linear equation. Multiplication by the integrating factor

$$\exp \left[\int_{\eta_i}^{\eta} d\eta' a(\eta') \Gamma_{\text{rec}}(\eta') \right] \quad (96)$$

and integration gives the stated expression. $\square$

The theorem shows exactly where abundance information enters: the initial comoving anomaly load, the equilibrium source, and the repair rate. These are OPH microphysical outputs, not SPARC parameters.

18.8 Linear Relaxation Transfer

Proposition 0H, one-pole diagnostic. For fixed k , assume the scalar diagnostic equation

$$\delta'_A = -\theta_A + 3\Phi' - a\Gamma_{\text{rec}} q_A (\delta_A - B_A \delta_b). \quad (97)$$

Define

$$\Xi_A = a\Gamma_{\text{rec}} q_A. \quad (98)$$

Then

$$\begin{aligned} \delta_A(\eta, k) &= W_A(\eta, \eta_i) \delta_A(\eta_i, k) \\ &+ \int_{\eta_i}^{\eta} d\eta' W_A(\eta, \eta') [-\theta_A + 3\Phi' + \Xi_A B_A \delta_b]_{\eta', k}, \end{aligned} \quad (99)$$

with

$$W_A(\eta, \eta') = \exp \left[- \int_{\eta'}^{\eta} d\tilde{\eta} \Xi_A(\tilde{\eta}, k) \right]. \quad (100)$$

Proof. Move the damping term to the left:

$$\delta'_A + \Xi_A \delta_A = -\theta_A + 3\Phi' + \Xi_A B_A \delta_b. \quad (101)$$

The integrating-factor solution is the displayed expression. $\square$

This formula has two useful limits. If $\Xi_A/\mathcal{H} \ll 1$, the anomaly evolves like pressureless free-falling matter. If $\Xi_A/\mathcal{H} \gg 1$, it tracks the equilibrium source up to derivative corrections. A common OPH-R1 operator for clusters and galaxies is work in progress.

18.9 Static-To-Linear Separation

Theorem 0I. The nonlinear static RAR law

$$g_{\text{obs}} = \frac{g_b}{1 - \exp[-\lambda_{\text{collar}} \sqrt{g_b/a_{0,\text{OPH}}}] \quad (102)$$

does not define the FLRW anomaly abundance or the linear CMB kernel by itself.

Proof. The RAR law is written for a static weak-field acceleration vector sourced by a resolved baryonic inhomogeneity. An exactly homogeneous FLRW background has no preferred spatial acceleration vector. Its scalar data are background densities and perturbation kernels, with no radial acceleration profile. In addition, the RAR map contains $\sqrt{g_b}$, so its first derivative is singular at $g_b = 0$ if treated as a homogeneous-background Taylor series. Therefore the static law cannot be linearized into CMB equations without an OPH parent functional. A scalar parent functional must emit $\rho_A(a)$ and $B_A(k, a)$, which belongs to the source-candidate layer. Physical Boltzmann promotion requires the finite covariant packet parent above, its source-provenance certificate, and pooled global reducers. $\square$

18.10 Parent Collar Functional

Scalar source target 0J. Let $\mathcal{C}_x(a)$ be the finite collar family contributing to a coarse FLRW cell x at scale factor a . Assume this finite-collar state selection has a regular FLRW limit. Let $\omega[\rho_b]$ be the OPH collar state selected by the baryonic source, and let R_C^{src} be the typed source routed by the source-route theorem:

$$R_C^{\text{src}}[\omega] = \begin{cases} \delta\langle K_C^{\text{anom}}[\sigma] \rangle, & \text{FIXED_REFERENCE_MODULAR_ENERGY,} \\ I_\omega(A : D|B), & \text{NONLINEAR_CMI_STRESS with matching receipt.} \end{cases} \quad (103)$$

Define

$$\rho_{A,\text{eq}}(x, a) c^2 = \frac{15}{8\pi^2 \ell(a)^4} \int_{\mathcal{C}_x(a)} d\mu_C R_C^{\text{src}}[\omega[\rho_b]]. \quad (104)$$

Then (104) is a scalar parent collar functional that emits a background equilibrium source and a linear response operator:

$$\bar{\rho}_{A,\text{eq}}(a) = \rho_{A,\text{eq}}[\bar{\rho}_b](a), \quad (105)$$

$$(K_{A,r}^{(\rho)})_{jj'}(a) = \left\langle v_{rj}, \frac{\delta \rho_{A,\text{eq}}(x, a)}{\delta \rho_b(x', a)} \bigg|_{\bar{\rho}_b} v_{rj'} \right\rangle, \quad (B_{A,r})_{jj'}(a) = \frac{\rho_b(a)}{\rho_{A,\text{eq}}(a)} (K_{A,r}^{(\rho)})_{jj'}(a). \quad (106)$$

Conditional argument. The collar measure, the conditional mutual information, and the selected collar state are quotient-local scalar data. The integral is therefore a scalar density on the coarse FLRW cell. Evaluating the functional on a homogeneous baryonic density gives the background equilibrium source. Taking the first functional derivative around that homogeneous state gives a translation-invariant and rotation-invariant linear response kernel. Its finite mode-basis matrix is $K_{A,r}^{(\rho)}(a)$, and division by the equilibrium background density gives $B_{A,r}(a)$. The scalar $B_A(k, a)$ is the diagonal, mode-decoupled special case after the commutator and off-diagonal leakage receipts pass. $\square$

The scalar reduction is accepted only on the physical projectors supplied by the scale bridge. In projector notation,

$$K_{A,r;IJ}^{(\rho)}(a) = P_{r,I} \mathcal{K}_{A,r}^{(\rho)}(a) P_{r,J}, \quad B_{A,r;IJ}(a) = \frac{\bar{\rho}_b(a)}{\bar{\rho}_{A,\text{eq}}(a)} K_{A,r;IJ}^{(\rho)}(a).$$

Reporting a scalar $B_A(k, a)$ requires

$$\epsilon_{\text{off}}(a) = \frac{\left(\sum_{I \neq J} |P_{r,I} \mathcal{K}_A P_{r,J}|^2 \right)^{1/2}}{|\mathcal{K}_A| + \epsilon_0} \leq \bar{\epsilon}_{\text{off}}$$

and, inside every degenerate projector,

$$P_{r,I} \mathcal{K}_A P_{r,I} = b_I(a) P_{r,I} + E_I, \quad \frac{|E_I|}{|b_I| + \epsilon_0} \leq \bar{\epsilon}_{\text{iso}}.$$

The evidence bundle records `mode_basis_hash`, `background_hash`, `clock_hash`, and `scale_bridge_hash`. $\rho_A(a)$, $B_A(k, a)$, and $\Gamma_{\text{rec}}(k, a)$ must share one frozen geometry and clock generation. A transition matrix eigenvalue per repair cycle is a diagnostic without a physical interval $\Delta\tau_r$, an active response block, and either a spectral-radius formula or a Lyapunov/semigroup bound are certified. A scalar $B_A(k, a)$ fails when off-diagonal mode mixing exceeds its frozen tolerance or when the result depends on unitary rotations inside a degenerate physical projector.

The finite-collar evaluation of (104) is the mathematical target for replacing a fitted dark-matter abundance, but it does not determine the full fluid closure. The scalar parent leaves open w_A , $c_{s,A}^2$, σ_A , Q_A^μ , exchange-current closure, a certified Γ_{rec} , recipient stress, local-frame/quotient covariance, gauge-invariant perturbations, and causal response. Those data must come from the finite covariant packet parent before the scalar tables can feed a physical Boltzmann module.

Proof obligation. Evaluate the collar measure and selected state from finite OPH screen/collar data in the homogeneous, perturbative, galaxy, and cluster regimes; lift the result to a finite covariant packet parent; prove exact stress-energy closure, gauge independence of $B_{A,r}$, common mode-basis compatibility with the primordial source, physical-time calibration for Γ_{rec} , recipient stress for nonzero exchange, exchange-current closure, local-frame and carrier-quotient invariance, causal/stable response, refinement convergence, and CDM-limit recovery.

18.11 Finite Repair Matrix

Theorem 0K. Let S be the finite packet-state space declared by one fixed-cutoff overlap interface. Let $\pi_{\text{eq}}(s|\rho_b)$ be the equilibrium packet distribution emitted by the parent collar functional. Let

$q(s, s')$ be an inverse-closed local proposal rule generated by the declared repair menu, with $q(s, s') > 0$ iff $q(s', s) > 0$. Define, for $s \neq s'$,

$$K(s, s') = q(s, s') \min\left(1, \frac{\pi_{\text{eq}}(s'|\rho_b)q(s', s)}{\pi_{\text{eq}}(s|\rho_b)q(s, s')}\right), \quad (107)$$

and

$$K(s, s) = 1 - \sum_{s' \neq s} K(s, s'). \quad (108)$$

Then K is a finite stochastic repair transition matrix with stationary distribution π_{eq} . If the proposal graph is connected and some rejection or hold probability is nonzero, K is irreducible and aperiodic.

Proof. Each row sums to one by construction, so K is stochastic. For $s \neq s'$, inverse-closed proposals make both directions available. Hence

$$\pi_{\text{eq}}(s)K(s, s') = \min(\pi_{\text{eq}}(s)q(s, s'), \pi_{\text{eq}}(s')q(s', s)) = \pi_{\text{eq}}(s')K(s', s). \quad (109)$$

Detailed balance proves stationarity. Connected proposal support gives irreducibility, and a nonzero hold probability gives aperiodicity. $\square$

This construction declares K from OPH data: packet states, local proposals, and the parent equilibrium distribution.

Scalar-load diagnostic. The executable packet diagnostic takes

$$S = \{0, 1, \dots, 40\}, \quad \pi_{\text{eq}}(n) = \text{Pois}(n; 5.363470441)$$

with truncation at 40, nearest-neighbor inverse-closed proposals, and hold probability 0.25. The resulting matrix has row-sum error 1.110×10^{-16} , detailed-balance error 6.939×10^{-18} , and

$$\gamma_{\text{repair step}} = 0.050717471.$$

This diagnostic is a finite packet schema. The OPH task is to derive that schema, or its replacement, from the finite collar states and then attach the stress, recipient, response, convergence, and frozen-likelihood receipts needed for physical transfer.

18.12 Finite Repair-Graph Gap

Theorem 0L. For a declared fixed-cutoff repair graph with transition matrix K , the relaxation time used by clusters and perturbations is computable from the second eigenvalue. If K is reversible with stationary distribution π_{eq} , then

$$r_\star = \max_{\lambda_j \neq 1} |\lambda_j(K)|, \quad g_{\text{abs}} = 1 - r_\star \quad (110)$$

in discrete repair-step units. If the commit interval is Δt_{commit} , the exact physical e-folding rate is

$$\Gamma_{\text{phys}} = -\frac{\log r_\star}{\Delta t_{\text{commit}}}, \quad \tau_{\text{phys}} = \frac{\Delta t_{\text{commit}}}{-\log r_\star}. \quad (111)$$

The expression $\Delta t_{\text{commit}}/(1 - r_\star)$ is only the small-gap approximation. For continuous-time generator L , the rate is the smallest positive eigenvalue of $-L$ on the mean-zero subspace. On a nonequilibrium nonnormal branch, eigenvalues alone are not enough; the parent must instead provide a Lyapunov, singular-value, or logarithmic-norm bound that excludes transient amplification.

Proof. Reversibility makes K self-adjoint in $L^2(\pi_{\text{eq}})$. The stationary vector has eigenvalue one. Every mean-zero mode decomposes into eigenvectors and decays by its eigenvalue per repair step. The slowest decaying nonstationary mode is therefore the largest nontrivial absolute eigenvalue $r_\star$. After $n = t/\Delta t_{\text{commit}}$ commits, its amplitude is $r_\star^n = \exp[-t(-\log r_\star)/\Delta t_{\text{commit}}]$, which gives the stated physical rate. The continuous-time statement follows by exponentiating the generator. $\square$

This is an exact finite-matrix target for the repair/commit microphysics. A cluster timescale is a prediction only after this matrix or its OPH state functional is supplied.

18.13 Jacobi Rate Family

For a real symmetric trace-free tidal tensor, rotational invariants are generated by

$$I_2 = \text{tr}(E^2), \quad I_3 = \text{tr}(E^3).$$

Every homogeneous rotationally invariant scalar rate built only from E_{ij} , without another dimensional scale, has the form

$$\Gamma_E = I_2^{1/4} F\left(\frac{I_3}{I_2^{3/2}}\right). \quad (112)$$

Rotational invariance and dimensional analysis do not select a unique rate.

The constant choice $F = 6^{-1/4}$ gives

$$\Gamma_J = (E_{ij}E^{ij}/6)^{1/4},$$

and reduces to $\sqrt{GM/r^3}$ for a point source. Its arithmetic and timescale tables are diagnostics, not a repair-clock theorem. Emission of the full invariant function from the finite repair generator and operational observer clock is work in progress; target observations cannot select it.

18.14 Homogeneous Anomaly State Selection

Theorem target 0N. Under capacity-saturated FLRW state selection, the homogeneous closure relation is

$$\Omega_{A0} + \Omega_{K0} = 1 - \Omega_{\Lambda, \text{OPH}} - \Omega_{b0} - \Omega_{\nu 0} - \Omega_{r0}, \quad \Omega_{\Lambda, \text{OPH}} = \left(\frac{H_{\text{dS}}}{H_0}\right)^2. \quad (113)$$

If $\Omega_{K0} = 0$ is separately selected or explicitly assumed, this reduces to the flat benchmark value for Ω_{A0} . Otherwise it is only a sum rule. If the homogeneous anomaly charge is transported as dust with no homogeneous repair exchange, then

$$\rho_A(a) = \rho_{A0}a^{-3}. \quad (114)$$

Explicit premise. The homogeneous state is capacity-saturated. The flat version additionally assumes or derives $\Omega_{K0} = 0$. A residual Friedmann constraint after H_0 , baryons, neutrinos, and radiation are supplied does not by itself select either the curvature sector or the anomaly abundance.

Conditional argument. The flat Friedmann equation at $a = 1$ is

$$1 = \Omega_{\Lambda, \text{OPH}} + \Omega_{b0} + \Omega_{\nu 0} + \Omega_{r0} + \Omega_{A0} + \Omega_{K0}. \quad (115)$$

Solving gives only $\Omega_{A0} + \Omega_{K0}$. Pressureless homogeneous transport with zero homogeneous exchange obeys

$$\rho'_A + 3\mathcal{H}\rho_A = 0,$$

whose solution is $\rho_A(a) = \rho_{A0}a^{-3}$. $\square$

Proof obligation. Derive the conserved homogeneous information-defect charge Q_A by the source-only anomaly abundance selector of Section 18.15 and supply an independent curvature label, or declare the abundance as cosmological state data.

Failure consequence. The CMB dark abundance is conditional when Q_A lacks `RHO_A_SOURCE_RECEIPT`. A nonzero exchange branch is also conditional unless the same finite parent exposes the recipient stress and equal-and-opposite exchange current that balances the exchange.

Capacity boundary. The screen capacity fixes the dimensionless de Sitter capacity relation; the capacity is the SL-4 estimate read from measured Λ (`./docs/STRANGE_LOOP_PRINCIPLES.md`), and the SI de Sitter scale additionally uses the selected OPH scale certificate. After H_0 and that scale display are supplied, it fixes $\Omega_{\Lambda, \text{OPH}}$; Λ itself is on no OPH output list. It does not by itself fix the baryon density, the neutrino density, the radiation density, the curvature sector, or a conserved homogeneous anomaly charge. A standalone OPH cosmology needs a theorem of the form

$$Q_A \equiv \mathcal{E}_{A,r} = F_{A,r}(Q_{A,r}^{\text{rel}}, \mu_{A,r}^{\text{rel}}, P_\star, N_{\text{CRC}}, \mathfrak{R}_{A,r}^{\text{rel}}, \mathcal{P}_r)$$

or an explicitly declared state-selection premise such as a flat benchmark using (61).

18.15 Source-Only Anomaly Abundance Selector

Definition 18.1 (C.52. Anomaly release state). *An anomaly release state at regulator r is*

$$\mathfrak{R}_{A,r}^{\text{rel}} = (\Sigma_r^{\text{rel}}, n_r^a, a_{\text{rel},r}, V_{\text{com},r}, \Pi_r^{(0)}, \mathfrak{B}_r^{\text{rel}}, h_{\text{rel}}, \mathfrak{L}_{\text{no data}}).$$

Here Σ_r^{rel} is the finite release hypersurface; n_r^a is its future unit normal in the parent tetrad; $a_{\text{rel},r}$ is the release scale factor; $V_{\text{com},r}$ is the declared comoving volume; $\Pi_r^{(0)}$ is the homogeneous FLRW zero-mode projector; and $\mathfrak{B}_r^{\text{rel}}$ is the release-branch boundary and sector data, including declared $P_\star$, N_{CRC} , compact-gauge branch, ordinary-sector branch, and exchange branch. The hash h_{rel} freezes the generation. The ledger $\mathfrak{L}_{\text{no data}}$ records the source dependency graph.

The release state is source-admissible only if all of its fields are fixed before CMB, BAO, supernova, weak-lensing, RSD, SPARC, cluster, and other likelihood comparisons.

Definition 18.2 (C.53. Finite anomaly load observable). *Let*

$$\mathcal{P}_r[X, g]$$

be the finite covariant collar-parent, or its source-side pre-load parent before total homogeneous amplitude selection. For a quotient state $q \in Q_{A,r}^{\text{rel}}$, define the local anomaly energy density on the release hypersurface by

$$\varepsilon_{A,r}(c; q) := T_{A,r}^{ab}(c; q) n_a n_b.$$

On the BW source route this readout has the form

$$\varepsilon_{A,r}(c; q) = \frac{15\hbar c}{8\pi^2} \frac{R_{A,r}^{\text{src}}(c, n, \ell; q)}{\ell^4} + \Delta_{A,r}^{\text{BW}}(c; q),$$

where $\Delta_{A,r}^{\text{BW}}$ is the finite-ball, curvature, gradient, localization, tomography, and variational/moment residual package required by the finite parent receipt.

The finite anomaly load observable is

$$\mathbb{L}_{A,r}(q) := \Pi_r^{(0)} \sum_{c \in \Sigma_r^{\text{rel}}(q)} V_{c,r}^{\text{phys}}(q) \varepsilon_{A,r}(c; q).$$

It has units of energy. It is defined on the physical quotient. Hidden representatives are not valid selector inputs.

The abundance selector is circular and invalid if $\mathbb{L}_{A,r}$ is computed from a parent occupation field whose normalization was set by an externally supplied Ω_A , ρ_{A0} , Planck residual, SPARC fit, cluster fit, Boltzmann residual, or likelihood comparison.

Definition 18.3 (C.54. Source-only anomaly abundance selector). *Let $Q_{A,r}^{\text{rel}}$ be the release quotient state space and let $\mu_{A,r}^{\text{rel}}$ be the declared OPH release law on that quotient. The source-only anomaly abundance selector is the deterministic functional*

$$F_{A,r}(Q_{A,r}^{\text{rel}}, \mu_{A,r}^{\text{rel}}, P_\star, N_{\text{CRC}}, \mathfrak{R}_{A,r}^{\text{rel}}, \mathcal{P}_r) := \mathbb{E}_{\mu_{A,r}^{\text{rel}}}[\mathbb{L}_{A,r}].$$

Equivalently,

$$\mathcal{E}_{A,r} = \sum_{q \in Q_{A,r}^{\text{rel}}} \mu_{A,r}^{\text{rel}}(q) \mathbb{L}_{A,r}(q).$$

The emitted homogeneous anomaly density is

$$\bar{\rho}_{A,r}(a) = \frac{\mathcal{E}_{A,r}}{a^3 V_{\text{com},r}}$$

on the conserved dust branch.

Definition 18.4 (C.55. Source MaxEnt release law). *The release law $\mu_{A,r}^{\text{rel}}$ is source-admissible if it is either*

$$\mu_{A,r}^{\text{rel}} = \mu_r|_{Q_{A,r}^{\text{rel}}}$$

renormalized on the release branch, where μ_r is a certified OPH quotient law, or the finite MaxEnt law

$$\mu_{A,r}^{\text{rel}}(q) = \frac{m_{A,r}^{\text{rel}}(q) \exp[-S_{A,r}^{\text{rel}}(q) - \sum_j \theta_j F_{j,r}^{\text{rel}}(q)]}{Z_{A,r}^{\text{rel}}}$$

on $Q_{A,r}^{\text{rel}}$, where every constraint $F_{j,r}^{\text{rel}}$, target $c_{j,r}$, unit, support, zero-mode treatment, sector convention, and refinement map is declared in the source constraint ledger.

No simulator output, Boltzmann residual, likelihood value, observational target, or tuned visual diagnostic may appear in $m_{A,r}^{\text{rel}}$, $S_{A,r}^{\text{rel}}$, $F_{j,r}^{\text{rel}}$, $c_{j,r}$, or the branch-selection ledger.

Theorem 18.5 (C.56. Source-only anomaly abundance selection). *Assume:*

1. $Q_{A,r}^{\text{rel}}$, $\mu_{A,r}^{\text{rel}}$, $P_\star$, N_{CRC} , $\mathfrak{R}_{A,r}^{\text{rel}}$, and $\mathcal{P}_r$ are source-only artifacts.
2. $\mathbb{L}_{A,r}$ factors through the physical quotient.
3. The local BW/source route, tomography, finite parent, stress closure, gauge, and refinement receipts required by the parent pass.

4. $\mu_{A,r}^{\text{rel}}$ is a certified OPH quotient law or a source MaxEnt law.
5. The no-data-use ledger excludes all observational likelihoods and all diagnostic solver outputs entering the prediction evaluation.

Then

$$\mathcal{E}_{A,r} = F_{A,r}(Q_{A,r}^{\text{rel}}, \mu_{A,r}^{\text{rel}}, P_{\star}, N_{\text{CRC}}, \mathfrak{R}_{A,r}^{\text{rel}}, \mathcal{P}_r)$$

is a source-only OPH anomaly abundance artifact. If the conserved dust branch holds, then

$$\bar{\rho}_{A,r}(a) = \frac{\mathcal{E}_{A,r}}{a^3 V_{\text{com},r}}$$

is a source-only homogeneous anomaly density.

Proof. The load observable $L_{A,r}$ is constructed from the parent-emitted anomaly stress or from the BW-normalized source route followed by stress tomography. By assumption it factors through $Q_{A,r}^{\text{rel}}$, so hidden carrier coordinates, port labels, repair schedules, gauge representatives, and worker metadata do not change it.

The release law $\mu_{A,r}^{\text{rel}}$ is either a certified OPH quotient law or the finite MaxEnt law determined by source constraints. Hence the expectation

$$\mathbb{E}_{\mu_{A,r}^{\text{rel}}}[L_{A,r}]$$

is a deterministic finite functional of source artifacts.

No observational likelihood or diagnostic solver output enters the source DAG by assumption. Therefore the artifact is source-only. On the conserved dust branch, the background parent equation gives

$$(a^3 \bar{\rho}_A V_{\text{com}})' = 0,$$

so the selected release load transports as

$$\bar{\rho}_A(a) = \mathcal{E}_{A,r} / (a^3 V_{\text{com}}).$$

□

Theorem 18.6 (C.57. No-data-use proof for $F_{A,r}$). *Let G_A^{load} be the source DAG whose terminal node is $\mathcal{E}_{A,r}$. Suppose every node verifier recomputes its predicate from committed artifacts, theorem witnesses, parent predicates, dependency manifests, and error envelopes, and suppose every root node is one of*

$$Q_{A,r}^{\text{rel}}, \quad m_{A,r}^{\text{rel}}, \quad S_{A,r}^{\text{rel}}, \quad F_{j,r}^{\text{rel}}, \quad c_{j,r}, \quad P_{\star}, \quad N_{\text{CRC}}, \quad \mathfrak{R}_{A,r}^{\text{rel}}, \quad \mathcal{P}_r$$

or a standard non-CMB physical constant allowed by the cosmology contract.

If the dependency ledger contains no CMB, BAO, supernova, weak-lensing, RSD, SPARC, cluster, likelihood, posterior, nuisance-fit, simulator residual, or diagnostic visual-comparison input, then

ANOMALY_ABUNDANCE_SOURCE_RECEIPT

passes.

Proof. Topologically order G_A^{load} . Root nodes are source-only by hypothesis. For a non-root node, the verifier accepts only if every parent has passed and the node value is recomputed from its declared parents and theorem witness. Induction over the DAG proves that the terminal value $\mathcal{E}_{A,r}$ is a deterministic function of source-only inputs. Since the ledger excludes every observational or likelihood target used for evaluation, no target-leakage path exists. A producer Boolean is not a proof because it is not a recomputation from the committed artifacts. □

Theorem 18.7 (C.58. Exact refinement compatibility of the abundance selector). *Let $s \succeq r$ and let*

$$c_{sr} : Q_{A,s}^{\text{rel}} \rightarrow Q_{A,r}^{\text{rel}}$$

be the physical coarse map. Suppose:

1. *the release laws are exactly compatible:*

$$(c_{sr})_{\#} \mu_{A,s}^{\text{rel}} = \mu_{A,r}^{\text{rel}};$$

2. *the load observable is exactly natural:*

$$\mathbb{L}_{A,s}(q_s) = \mathbb{L}_{A,r}(c_{sr}q_s) \quad \forall q_s \in Q_{A,s}^{\text{rel}};$$

3. *the release volume and homogeneous projector commute with refinement.*

Then

$$\boxed{\mathcal{E}_{A,s} = \mathcal{E}_{A,r}.}$$

Proof. By definition,

$$\mathcal{E}_{A,s} = \sum_{q_s} \mu_{A,s}^{\text{rel}}(q_s) \mathbb{L}_{A,s}(q_s).$$

Using load naturality,

$$\mathcal{E}_{A,s} = \sum_{q_s} \mu_{A,s}^{\text{rel}}(q_s) \mathbb{L}_{A,r}(c_{sr}q_s).$$

Push the fine law forward along c_{sr} . Exact law compatibility gives

$$\sum_{q_s : c_{sr}q_s = q_r} \mu_{A,s}^{\text{rel}}(q_s) = \mu_{A,r}^{\text{rel}}(q_r).$$

Therefore

$$\mathcal{E}_{A,s} = \sum_{q_r} \mu_{A,r}^{\text{rel}}(q_r) \mathbb{L}_{A,r}(q_r) = \mathcal{E}_{A,r}.$$

□

Theorem 18.8 (C.59. Approximate refinement bound for the abundance selector). *Let $s \succeq r$. Suppose*

$$\left\| (c_{sr})_{\#} \mu_{A,s}^{\text{rel}} - \mu_{A,r}^{\text{rel}} \right\|_{\text{TV}} \leq \delta_{sr}^{\mu},$$

and

$$\sup_{q_s} |\mathbb{L}_{A,s}(q_s) - \mathbb{L}_{A,r}(c_{sr}q_s)| \leq \delta_{sr}^L.$$

Let

$$L_{\max,r} := \|\mathbb{L}_{A,r}\|_{\infty}.$$

Then

$$\boxed{|\mathcal{E}_{A,s} - \mathcal{E}_{A,r}| \leq \delta_{sr}^L + 2L_{\max,r} \delta_{sr}^{\mu}.}$$

For a refinement chain $r = r_0 \preceq r_1 \preceq \dots \preceq r_n = s$,

$$|\mathcal{E}_{A,s} - \mathcal{E}_{A,r}| \leq \sum_{k=0}^{n-1} \left(\delta_{r_{k+1}r_k}^L + 2L_{\max,r_k} \delta_{r_{k+1}r_k}^{\mu} \right).$$

If the right-hand side tends to zero along a cofinal family, the selected anomaly load has a regulator-independent limit.

Proof. Insert and subtract the coarse load pulled back to the fine law:

$$|\mathcal{E}_{A,s} - \mathcal{E}_{A,r}| \leq |\mathbb{E}_{\mu_s}[\mathbb{L}_{A,s} - \mathbb{L}_{A,r} \circ c_{sr}]| \\ + |\mathbb{E}_{\mu_s}[\mathbb{L}_{A,r} \circ c_{sr}] - \mathbb{E}_{\mu_r}[\mathbb{L}_{A,r}]|.$$

The first term is at most δ_{sr}^L . The second term is bounded by the total-variation estimate

$$2\|\mathbb{L}_{A,r}\|_\infty \delta_{sr}^\mu.$$

Summing the one-step bounds gives the chain estimate. Cofinal convergence follows immediately. $\square$

Theorem 18.9 (C.60. Source-only ρ_A promotion). *The receipt*

$$RHO_A_SOURCE_RECEIPT$$

passes iff

$$RHO_A_TRANSPORT_RECEIPT \wedge ANOMALY_ABUNDANCE_SOURCE_RECEIPT$$

passes under the same frozen generation, unit ledger, release state, and parent generation. When this receipt passes, $\bar{\rho}_A(a)$ may be labelled

$$SOURCE_ONLY_ANOMALY_ABUNDANCE.$$

If only the transport receipt passes, the label is

$$PHYSICAL_PARENT_WITH_CONDITIONAL_ABUNDANCE.$$

Proof. The transport receipt proves that the finite parent evolves a homogeneous anomaly load consistently once that load is supplied. The abundance source receipt proves that the load itself is selected by the OPH source functional $F_{A,r}$, without target leakage and with refinement compatibility. Their conjunction is exactly the statement that ρ_A is both physically transported and source-selected. Without the selector, the amplitude is external state data. $\square$

The closed paper-side abundance receipt is

$$ANOMALY_ABUNDANCE_SOURCE_RECEIPT = ANOMALY_RELEASE_STATE_RECEIPT \wedge ANOMALY_LOAD_OBSERVABLE_RECEIPT \\ \wedge SOURCE_MAXENT_RELEASE_LAW_RECEIPT \wedge LOAD_QUOTIENT_INVARIANCE_RECEIPT \\ \wedge LOAD_NO_CIRCULARITY_RECEIPT \wedge LOAD_NO_DATA_USE_RECEIPT \\ \wedge LOAD_REFINEMENT_COMPATIBILITY_RECEIPT.$$

Then

$$RHO_A_SOURCE_RECEIPT = RHO_A_TRANSPORT_RECEIPT \wedge ANOMALY_ABUNDANCE_SOURCE_RECEIPT.$$

The source-ready dark Boltzmann handoff is

$$DARK_PHYSICAL_BOLTZMANN_SOURCE_RECEIPT = FINITE_COVARIANT_PARENT_RECEIPT_V2 \wedge PHYSICAL_ANOMALY_KINETICS_RECEIPT \\ \wedge RHO_A_TRANSPORT_RECEIPT \wedge FROZEN_GENERATION_MANIFEST_RECEIPT \\ \wedge BOLTZMANN_INITIAL_CONSTRAINT_RECEIPT \wedge BOLTZMANN_ADAPTATION_RECEIPT \\ \wedge CDM_LIMIT_RECOVERY_RECEIPT.$$

For a source-only dark abundance claim,

$$\text{SOURCE_ONLY_ANOMALY_ABUNDANCE} \iff \text{RHO_A_SOURCE_RECEIPT}.$$

Equivalently, the selector computes

$$F_{A,r} = \mathbb{E}_{\mu_{A,r}^{\text{rel}}} \left[\Pi_r^{(0)} \sum_{c \in \Sigma_r^{\text{rel}}} V_{c,r}^{\text{phys}} T_{A,r}^{ab}(c) n_a n_b \right].$$

On the BW source branch,

$$F_{A,r} = \mathbb{E}_{\mu_{A,r}^{\text{rel}}} \left[\Pi_r^{(0)} \sum_{c \in \Sigma_r^{\text{rel}}} V_{c,r}^{\text{phys}} \left(\frac{15\hbar c}{8\pi^2} \frac{R_{A,r}^{\text{src}}(c, n, \ell)}{\ell^4} + \Delta_{A,r}^{\text{BW}}(c) \right) \right].$$

The selector integrates the finite source stress over the release hypersurface and averages it over the declared source quotient ensemble. It does not read a Friedmann residual, a Planck fit, a SPARC fit, a cluster fit, or a Boltzmann likelihood.

19 Falsifiers

The dark continuation fails under any of the following results:

1. finite collar data do not emit a covariant response parent without using the target observations;
2. the source-derived static limit fails full conservative disk solutions;
3. one frozen parent cannot satisfy galaxy and Cassini likelihoods;
4. the matched C_i statistic has a significant corrected population offset;
5. a frozen wide-binary or satellite likelihood rejects the parent;
6. the parent lacks causal stability, total exchange-current closure, or acceptable propagation and preferred-frame behavior;
7. galaxy or cluster lensing conflicts with the action-derived metric potentials;
8. one source-selected abundance and perturbation model cannot fit CMB, CMB lensing, BAO, RSD, weak lensing, and growth;
9. RAR redshift evolution conflicts with the registered $a_{0,\text{app}}(z)$;
10. a positive transported density is equated to the signed QUMOND source without an internal-stress or metric-response theorem.

Cassini excludes the universal static QUMOND continuation. This result does not falsify the recovered OPH core.

20 Reproducibility

The static and timing diagnostics are reproduced by:

```
python3 code/dark_matter/scripts/sparc_rar_compare.py
python3 code/dark_matter/scripts/sparc_rotation_curve_compare.py
python3 code/dark_matter/scripts/sparc_systematic_likelihood.py
python3 code/dark_matter/scripts/dark_repair_transition_matrix.py
python3 code/dark_matter/scripts/dark_jacobi_repair_clock.py
```

The SPARC data files used are public measurement tables from the SPARC page [7].

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